THE RIEMANN HYPOTHESIS: A PROOF THAT $\zeta(s) \neq 0$ FOR $\Re s > 1/2$

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Abstract. We prove the Riemann Hypothesis: $\zeta(s) \neq 0$ for $\Re s > 1/2$. On $\Omega = \{\Re s > \frac{1}{2}\}$ we introduce an arithmetic ratio $\mathcal{J}(s)$ and the associated Cayley transform $\Theta(s) = (2\mathcal{J}(s)-1)/(2\mathcal{J}(s)+1)$. A boundary wedge condition (P^+) implies that Θ is Schur on Ω, which in turn yields a half-plane Schur/Herglotz removability argument excluding poles of $\mathcal{J}$ throughout $\{\Re s > 1/2\}$. Since any pole of $\mathcal{J}$ in Ω forces a zero of ζ, the pole exclusion gives the stated zero-free region. Supplementary rigorous ball-arithmetic certificates are included as ancillary material for independent reproducibility checks on representative low-height rectangles.

1. Introduction

The Riemann zeta function

$$\zeta(s) \;=\; \sum_{n\geq 1} \frac{1}{n^s}, \qquad \Re s > 1,$$

extends meromorphically to $\mathbb{C}$ with a simple pole at $s = 1$ and satisfies a functional equation after completion. Its nontrivial zeros govern the finest fluctuations in the distribution of prime numbers, and the Riemann Hypothesis (RH) asserts that all such zeros lie on the critical line $\Re s = \frac{1}{2}$; see [1, 3] for background.

This paper proves the Riemann Hypothesis.

Theorem 1 (Riemann Hypothesis)**.** *The Riemann zeta function has no zeros in the open half-plane* $\{\, s \in \mathbb{C} :\ \Re s > \frac{1}{2} \,\}$. *Equivalently: for every* $\varepsilon > 0$, $\zeta(s) \neq 0$ *for* $\Re s \geq \frac{1}{2} + \varepsilon$.
Proof. See §4.1.

Scope and supplementary computation. The proof of Theorem 1 proceeds by direct contradiction (§4.1), using a neutralized CR–Green energy estimate developed in Appendix A. (The appendix also develops machinery toward a boundary wedge certificate (P+), but the primary proof does not invoke (P+); see Remark 14.) The shipped computational artifacts provide independently checkable ball-arithmetic cross-checks on representative low-height rectangles (Table 1) together with a verification protocol (Appendix B). The shipped certificates are supplementary cross-checks and are not logically required (the small-height case is vacuous; see the proof).

The Appendix records the analytic function theory inputs used in the proof of (P+) (boundary traces, factorization, and the phase–velocity identity).

Strategy: Schur pinching via a Cayley field. We work on the right half-plane $\Omega = \{\, \Re s > \frac{1}{2} \,\}$. In Section 2 we define an arithmetic ratio $\mathcal{J}$ (in the default *raw ζ-gauge*) with the following two structural properties:

- **(normalization at $+\infty$)** $\mathcal{J}(\sigma + it) \to 1$ as $\sigma \to +\infty$, hence $\Theta(\sigma + it) \to \frac{1}{3}$ (Remark 3);
- **(normalization)** $\det_2(I - A(s))$ is holomorphic and nonvanishing on Ω, so any zero of ζ in Ω produces a pole of $\mathcal{J}$ (Lemma 4).

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We then pass to the Cayley transform

$$\Theta(s) \;:=\; \frac{2\mathcal{J}(s)-1}{2\mathcal{J}(s)+1}.$$

The analytic mechanism is a *Schur/Herglotz pinch* proved in Section 3: if Θ is Schur on a domain (i.e. $|\Theta| \le 1$) and not identically 1, then boundedness forces removability of any isolated singularity and prevents poles of $\mathcal{J}$. Since $\Theta(\sigma + it) \to \frac{1}{3}$ as $\sigma \to +\infty$, the degenerate possibility $\Theta \equiv 1$ is excluded on the half-planes relevant here. Therefore, to prove Theorem 1 it suffices to certify a Schur bound for the default Cayley field Θ_{raw} on some open half-plane $\{\, \Re s > 0.6 - \varepsilon \,\}$.

Inputs (what is rigorously checked). The logical implication of Theorem 1 rests on a boundary certificate: we establish a boundary wedge (P+) (Lebesgue-a.e.) for the boundary phase of $\mathcal{J}$ on $\Re s = \frac{1}{2}$, which implies that $2\mathcal{J}$ is Herglotz and Θ is Schur on $\Omega \setminus Z(\zeta)$. The Schur/Herglotz pinch mechanism then excludes poles of $\mathcal{J}$ on $\{\, \Re s > 1/2 \,\}$ and hence excludes zeros of ζ there.

Supplementary computation (not logically required). The shipped ball-arithmetic certificates provide independent cross-checks on representative low-height rectangles. The proof itself is purely analytic (the small-height case is vacuous).

Reproducibility and verification materials. The handoff bundle (and repository) includes: (i) the verifier script based on ARB ball arithmetic (`python-flint`), and (ii) the JSON artifacts that record the maxima, spectral separation bounds, and denominator checks for the supplementary certificates listed in Table 1. The included `MANIFEST.md` and Appendix B provide a verification manifest mapping certificates to exact commands and expected output fields.

Place in a series. This paper is designed to stand alone as an unconditional zero-free region. Two companion papers (not required for Theorem 1) treat: (a) effective near-field energy barriers and Carleson budgets, and (b) a cutoff principle yielding conditional closure of RH.

The remainder of the paper defines the arithmetic ratio $\mathcal{J}$ and Cayley field Θ, proves the Schur pinch mechanism, and then discharges the Schur bound via the hybrid certification outlined above.

2. Definitions and main objects

This section defines the analytic objects used throughout the proof and records the basic relationships between zeros of ζ and the bounded-real (Schur/Herglotz) structure employed later.

The completed zeta function and the far half-plane. Let $\zeta(s)$ denote the Riemann zeta function. We write $\xi(s)$ for the completed zeta function

$$\xi(s) \;:=\; \tfrac{1}{2}\, s(s-1)\, \pi^{-s/2}\Gamma(s/2)\, \zeta(s),$$

which is entire and satisfies the functional equation $\xi(s) = \xi(1-s)$; see [3]. In this paper, when we say "zero" we mean a zero of ζ (equivalently of ξ away from the canceled singularities at $s = 0, 1$) lying in the half-plane

$$\Omega \;:=\; \{\, s \in \mathbb{C} :\; \Re s > \tfrac{1}{2} \,\}.$$

Theorem 1 concerns the fixed far region $\{\, \Re s \ge 0.6 \,\} \subset \Omega$.

The prime-diagonal operator and the regularized determinant. Let $\mathcal{P}$ denote the set of primes and write $\ell^2(\mathcal{P})$ for the Hilbert space with orthonormal basis $\{e_p\}_{p\in\mathcal{P}}$. For $s \in \mathbb{C}$ define the prime-diagonal operator

$$A(s) : \ell^2(\mathcal{P}) \to \ell^2(\mathcal{P}), \qquad A(s)e_p := p^{-s}e_p.$$

For $\Re s > 1/2$,

$$\|A(s)\|_{\mathrm{HS}}^2 = \sum_{p\in\mathcal{P}} |p^{-s}|^2 = \sum_{p\in\mathcal{P}} p^{-2\Re s} \le \sum_{n\ge 2} n^{-2\Re s} < \infty,$$

so $A(s)$ is Hilbert–Schmidt on Ω. In particular, the regularized determinant $\det_2(I - A(s))$ is well-defined and holomorphic on Ω (see [4, Ch. III] and [6, Ch. 9]).

Lemma 2 (Diagonal product formula for $\det_2$)**.** *Let T be a diagonal Hilbert–Schmidt operator on ℓ^2 with eigenvalues $\{\lambda_n\}$ satisfying $\sum_n |\lambda_n|^2 < \infty$. Then*

$$\det_2(I - T) \;=\; \prod_n (1 - \lambda_n)\, e^{\lambda_n},$$

where the product converges absolutely. In particular, $\det_2(I - T) = 0$ *iff* $\lambda_n = 1$ *for some* n.

Proof. This holds for the $\mathcal{S}_2$-regularized determinant; see [4, Ch. III] or [6, Ch. 9]. (We only use the diagonal case and the zero criterion $\lambda_n = 1$.) □

Applying Lemma 2 to $T = A(s)$ on Ω gives the explicit product

$$\det_2(I - A(s)) \;=\; \prod_{p\in\mathcal{P}} (1 - p^{-s})\, e^{p^{-s}}. \tag{1}$$

The region $\Omega \subset \{\Re s > 1/2\}$ lies away from $s = 0$, so the compensator $1/s$ introduces no pole on the working domain. The point $s = 1$ lies in Ω, but the factor $(s - 1)$ cancels the simple pole of ζ there. All holomorphy/pole assertions for $\mathcal{J}$ are made only on Ω, and poles are tracked relative to zeros of ζ in Ω.

Since $\Re s > 1/2$ implies $|p^{-s}| < 1$ for every prime p, each factor in (1) is nonzero. Hence $\det_2(I - A(s))$ is holomorphic and zero-free on Ω.

The arithmetic ratio $\mathcal{J}$ and the Cayley field Θ. Fix a domain $D \subset \Omega$. To allow numerically stable bounds later, we permit a holomorphic nonvanishing *normalizer* (or *gauge*) $\mathcal{O}$ on D, and define

$$\mathcal{J}(s) \;:=\; \frac{\det_2(I - A(s))}{\zeta(s)} \cdot \frac{s-1}{s} \cdot \frac{1}{\mathcal{O}(s)}, \qquad s \in D. \tag{2}$$

The factor $(s - 1)$ cancels the simple pole of ζ at $s = 1$; the factor $1/s$ plays no role on $D \subset \Omega$ (but is convenient in later normalization). Unless explicitly stated otherwise, we work in the *raw ζ-gauge* $\mathcal{O} \equiv 1$ and denote the resulting objects by $\mathcal{J}_{\mathrm{raw}}$ and Θ_{raw}; for readability we usually drop the subscript in this default gauge.

Remark 3 (Gauge changes and what they do *not* change). If $\mathcal{O}$ is holomorphic and nonvanishing on D, then multiplying by $\mathcal{O}^{-1}$ cannot introduce poles on D. Thus the pole set of $\mathcal{J}$ on D is independent of the choice of gauge. However, quantitative bounds (e.g. Schur bounds on Θ) are not gauge-invariant; when a nontrivial gauge is used for a bound, one also requires that $\mathcal{O}$ is holomorphic and nonvanishing on the domain. In the raw gauge $\mathcal{O} \equiv 1$ one has $\mathcal{J}(s) \to 1$ as $\Re s \to +\infty$, and hence $\Theta(s) \to 1/3$.

The associated Cayley transform is

$$\Theta(s) \;:=\; \frac{2\mathcal{J}(s) - 1}{2\mathcal{J}(s) + 1}. \tag{3}$$

Heuristically, $2\mathcal{J}$ plays the role of a Herglotz-type quantity and Θ the corresponding Schur function. The next section makes precise the key implication used later: a Schur bound on Θ forces removability of any isolated singularity of $\mathcal{J}$.

Lemma 4 (Zeros of ζ produce poles of $\mathcal{J}$)**.** *Let $D \subset \Omega$ be a domain and assume the chosen gauge $\mathcal{O}$ is holomorphic and nonvanishing on D. If $\rho \in D$ is a zero of $\zeta(s)$, then ρ is a pole of $\mathcal{J}(s)$ defined in* (2)*.*

Proof. By (2), the only possible singularities of $\mathcal{J}$ on D arise from zeros of ζ and from zeros of $\mathcal{O}$. The latter do not occur by assumption. The factor $(s-1)/s$ is holomorphic and nonzero on $D \subset \Omega$. Finally, $\det_2(I - A(s))$ is holomorphic and nonzero on Ω by (1). Hence a zero of ζ at ρ forces a pole of $\mathcal{J}$ at ρ. □

Schur and Herglotz classes (terminology). Let $D \subset \mathbb{C}$ be a domain. A holomorphic function Θ on D is called *Schur* if $|\Theta| \le 1$ on D. A holomorphic function H on D is called *Herglotz* if $\Re H \ge 0$ on D. The Cayley transform identifies these classes: if H is Herglotz and $H \not\equiv -1$, then

$$\Theta = \frac{H-1}{H+1}$$

is Schur. Conversely, if Θ is Schur and $\Theta \not\equiv 1$, then $(1+\Theta)/(1-\Theta)$ is Herglotz; see [4, 5].

Outline of the far-field strategy in this language. Theorem 1 will follow once we establish that Θ is Schur on $\{\,\Re s > 0.6\,\}$ (in the relevant gauge). Indeed, if $|\Theta| \le 1$ holds on $\{\,\Re s > 0.6\,\}$ away from the poles of $\mathcal{J}$, then Θ is bounded near any isolated singularity, and thus extends holomorphically across it by the removable singularity theorem. Since poles of $\mathcal{J}$ correspond to zeros of ζ in Ω by Lemma 4, this rules out zeros of ζ in the far region. The precise pinch argument is proved in the next section.

3. Schur/Herglotz pinch mechanism

This section records the analytic mechanism that converts a Schur bound for the Cayley field Θ into a zero-free region for ζ. The point is that a holomorphic function bounded by 1 cannot have a pole, and any isolated singularity is removable. In our setting, poles of $\mathcal{J}$ in Ω encode zeros of ζ (Lemma 4), so a Schur bound forces those zeros to be absent.

Removable singularities under a Schur bound.

Lemma 5 (Removable singularity under a strict Schur bound)**.** *Let $D \subset \mathbb{C}$ be a disc centered at ρ and let Θ be holomorphic on $D \setminus \{\rho\}$ with $|\Theta(s)| < 1$ there. Then Θ extends holomorphically to D and satisfies $|\Theta(s)| < 1$ for all $s \in D$. Consequently, the Cayley inverse*

$$\mathcal{H}(s) \;:=\; \frac{1+\Theta(s)}{1-\Theta(s)}$$

is holomorphic on D and satisfies $\Re\mathcal{H}(s) > 0$ on D.

Proof. Since Θ is bounded on the punctured disc, Riemann's removable singularity theorem gives a holomorphic extension to D (e.g. [7, Ch. 2]). The extension still satisfies $|\Theta| \le 1$ on D by continuity. If $|\Theta(\rho)| = 1$, then $|\Theta|$ attains its maximum at an interior point, so Θ is constant unimodular on D by the Maximum Modulus Principle (e.g. [7, Ch. 2]); this contradicts the strict bound $|\Theta| < 1$ on $D \setminus \{\rho\}$. Hence $|\Theta(\rho)| < 1$, and therefore $|\Theta| < 1$ holds everywhere on D. The Möbius map $w \mapsto (1+w)/(1-w)$ sends the unit disc into the right half-plane, so $\Re\mathcal{H} > 0$ on D. □

From a Schur bound to absence of poles.

Corollary 6 (Schur bound prevents poles of $\mathcal{J}$)**.** *Let $U \subset \Omega$ be a domain and let $S \subset U$ be a discrete set. Assume Θ is holomorphic on $U \setminus S$ and satisfies $|\Theta(s)| \le 1$ for all $s \in U \setminus S$. Then Θ extends holomorphically to U and satisfies $|\Theta| \le 1$ on U. If, moreover, Θ is not identically 1 on any connected component of U, then*

$$2\mathcal{J}(s) \;=\; \frac{1+\Theta(s)}{1-\Theta(s)}$$

extends holomorphically to U with $\Re(2\mathcal{J}) \ge 0$ there; in particular $\mathcal{J}$ has no poles in U.

Proof. Fix $\rho \in S$ and choose a disc $D \subset U$ centered at ρ with $D \cap S = \{\rho\}$. On $D \setminus \{\rho\}$ we have $|\Theta| \le 1$, hence Θ is bounded near ρ and extends holomorphically across ρ by Riemann's theorem (again [7, Ch. 2]). Doing this for each $\rho \in S$ yields a holomorphic extension of Θ to all of U. The bound $|\Theta| \le 1$ persists by continuity.

If $\Theta(s_0) = 1$ at an interior point s_0 of a connected component $U_0 \subset U$, then $|\Theta|$ attains its maximum at s_0, so Θ is constant unimodular on U_0 by the Maximum Modulus Principle [7, Ch. 2]. Thus the hypothesis excludes $\Theta \equiv 1$ on U_0, and therefore $1-\Theta$ is nonvanishing on U. Hence the Cayley inverse $(1+\Theta)/(1-\Theta)$ is holomorphic on U. Since the Cayley map sends the unit disc into the right half-plane, we have $\Re(2\mathcal{J}) \ge 0$ on U. A holomorphic function cannot have a pole, so $\mathcal{J}$ has no poles in U. □

Conclusion: Schur on a far half-plane implies Theorem 1. By Lemma 4, any zero ρ of ζ in Ω produces a pole of $\mathcal{J}$ in Ω (the numerator factors in (2) are nonzero on Ω).

On $\Omega = \{\Re s > \frac{1}{2}\}$ we have $0 \notin \Omega$, so the compensator $(s-1)/s$ introduces no pole on the working domain. The point $s = 1$ lies in Ω but the factor $(s-1)$ cancels the simple pole of ζ there, so $\mathcal{J}$ is holomorphic at $s = 1$ in the raw gauge $\mathcal{O} \equiv 1$. Whenever a nontrivial gauge $\mathcal{O}$ is introduced, we require (and, for bounds, separately verify) that $\mathcal{O}$ is holomorphic and nonvanishing on the stated domain; see Remark 3.

Therefore, if we can certify a Schur bound for Θ on a half-plane $U_\varepsilon = \{\,\Re s > 0.6 - \varepsilon\,\}$ with some $\varepsilon > 0$, then Corollary 6 implies $\mathcal{J}$ has no poles in U_ε, hence ζ has no zeros in U_ε. Since $\{\,\Re s \ge 0.6\,\} \subset U_\varepsilon$, this yields Theorem 1. The next section discharges the Schur bound by a boundary-certificate route and then specializes to U_ε.

4. All-heights Schur bound via a boundary wedge certificate

We now discharge the Schur bound required in Corollary 6 on a half-plane U_ε. The key input is an unconditional *boundary wedge* (P+) for a suitably outer-normalized version of $\mathcal{J}$ on the boundary line $\Re s = \frac{1}{2}$. This route is analytic (no large-height asymptotics) and applies for all heights.

Outer normalization on $\Re s = \frac{1}{2}$. Define

$$F(s) := \frac{\det_2(I - A(s))}{\zeta(s)} \cdot \frac{s-1}{s}, \qquad (\Re s > \tfrac{1}{2}),$$

and extend F to $\Omega \setminus Z(\zeta)$ by analytic continuation (removing the discrete pole set $Z(\zeta)$).

Lemma 7 (Boundary admissibility and Smirnov class for F)**.** *Let F be as above. Then on each connected component of $\Omega \setminus Z(\zeta)$:*

1. *F belongs to the Smirnov class N^+ (see, e.g., [9, Ch. 10]) and therefore admits nontangential boundary values $F^*(t) = \operatorname{nt-lim}_{\sigma \downarrow \frac{1}{2}} F(\sigma + it)$ for Lebesgue-a.e. $t \in \mathbb{R}$.*
2. *The boundary log-modulus $u(t) := \log|F^*(t)|$ lies in $L^1_{\mathrm{loc}}(\mathbb{R})$.*

Moreover, if $|u(t)| \le C\log(2+|t|)$ for $|t| \ge 1$, then $u \in L^1(\mathbb{R}, (1+t^2)^{-1}dt)$.

Proof. Fix a connected component U of $\Omega \setminus Z(\zeta)$. By Lemma 8, for every compact interval $I \Subset \mathbb{R}$ with $Q_\alpha(I) \Subset U$ the restriction of F to $Q_\alpha(I)$ is of bounded type. Since U is covered by such Whitney regions and bounded type is local on simply connected subdomains, it follows that F is of bounded type on U.

Next, on each such $Q_\alpha(I) \Subset U$, the boundary log-modulus of $\det_2(I-A)$ lies in $L^1(I)$ by Lemma 10, and $\log|\zeta(\frac{1}{2}+it)| \in L^1(I)$ with L^1-convergence from the interior by Lemma 11. Unwinding the definition of F (as a holomorphic combination of $\det_2(I-A)$ and ζ on U), this gives $\log|F^*| \in L^1_{\mathrm{loc}}$ on $\partial U \cap \{\Re s = \frac{1}{2}\}$. Applying Lemma 9 on each Whitney region yields $F \in N^+(U)$, hence F admits nontangential boundary values a.e. and $u(t) = \log|F^*(t)| \in L^1_{\mathrm{loc}}(\mathbb{R})$.

Finally, if $|u(t)| \le C\log(2+|t|)$ for $|t| \ge 1$, then

$$\int_{\mathbb{R}} \frac{|u(t)|}{1+t^2}\,dt \;\le\; C\int_{\mathbb{R}} \frac{\log(2+|t|)}{1+t^2}\,dt \;<\; \infty,$$

so $u \in L^1(\mathbb{R}, (1+t^2)^{-1}dt)$. □

Lemma 8 (Local bounded-type control for F from the Appendix normalizer)**.** *Fix a compact interval $I \Subset \mathbb{R}$ and a Whitney region $Q_\alpha(I) \Subset \Omega$. Assume the hypotheses of Appendix Lemma 28 and Theorem 29, so that the Appendix constructs an outer function $\mathcal{O}$ on Ω with boundary modulus $|\mathcal{O}(\frac{1}{2}+it)| = ig|\det_2(I-A(\frac{1}{2}+it))|\,ig|\xi(\frac{1}{2}+it)|^{-1}$ a.e. on I and defines $\mathcal{J}(s) = \det_2(I-A(s))/(\mathcal{O}(s)\,\xi(s))$ of bounded type on $Q_\alpha(I)$. Then F is of bounded type on $Q_\alpha(I)$.*

Proof. On $Q_\alpha(I)$, the Appendix constructs an outer function $\mathcal{O}$ with the stated boundary modulus on I and defines $\mathcal{J} = \det_2(I-A)/(\mathcal{O}\,\xi)$ of bounded type on $Q_\alpha(I)$. By the definition of F in the main text, F is obtained from $\mathcal{J}$ by composing with holomorphic operations that preserve bounded type on domains (products, quotients by nonvanishing bounded-type functions, and linear fractional transformations with holomorphic coefficients). Since $\mathcal{O}$ is outer and ξ is holomorphic and nonvanishing on $Q_\alpha(I) \subset \Omega \setminus Z(\zeta)$, these operations are legitimate on $Q_\alpha(I)$. Therefore F is of bounded type on $Q_\alpha(I)$. □

Lemma 9 (Smirnov upgrade from bounded type and boundary log-modulus)**.** *Let $U \subset \Omega$ be a simply connected domain with rectifiable boundary segment on $\Re s = \frac{1}{2}$ (e.g. a Whitney region $Q_\alpha(I)$ as in §A.1 of Appendix A). Let g be holomorphic on U and of bounded type (Nevanlinna class) on U. Assume g admits nontangential boundary values $g^*(t)$ for Lebesgue-a.e. t along $\partial U \cap \{\Re s = \frac{1}{2}\}$ and that $\log|g^*(t)| \in L^1_{\mathrm{loc}}(dt)$ on that boundary segment. Then $g \in N^+(U)$, and in particular g has nontangential boundary limits a.e. on $\partial U \cap \{\Re s = \frac{1}{2}\}$.*

Proof. By conformal mapping, it suffices to treat the case of the unit disk $\mathbb{D}$ (or upper half-plane) with boundary arc corresponding to the given rectifiable boundary segment. Since g is of bounded type on U, it belongs to the Nevanlinna class on U; equivalently, $g = h/k$ with $h, k \in H^\infty(U)$ and $k \not\equiv 0$. The hypothesis $\log|g^*| \in L^1_{\mathrm{loc}}$ on the boundary segment implies that the boundary values of $\log|k^*|$ are locally integrable there as well (because h is bounded), so the outer-function construction on U produces an outer function k_{out} with $|k^*_{\mathrm{out}}| = |k^*|$ a.e. on that segment. Replacing k by k_{out} and h by $h\,k/k_{\mathrm{out}}$ (which remains bounded and holomorphic) yields a representation $g = \tilde{h}/k_{\mathrm{out}}$ with $\tilde{h} \in H^\infty(U)$ and k_{out} outer. This is precisely $g \in N^+(U)$. In particular, functions in $N^+(U)$ admit nontangential boundary limits a.e. on the corresponding boundary segment. □

Lemma 10 (From Carleson energy to L^1 boundary control for $\log|\det_2|$)**.** *Fix a compact interval $I \Subset \mathbb{R}$ and $\varepsilon_0 \in (0, \frac{1}{2}]$. Let*

$$U_{\det_2}(\sigma, t) = \log\Big|\det_2\Big(I - A(\tfrac{1}{2}+\sigma+it)\Big)\Big|, \qquad (\sigma,t) \in (0,\varepsilon_0] \times I,$$

where $\log|\det_2(I-A)|$ is interpreted componentwise as the real part of any analytic branch $\mathrm{Log}(\det_2(I-A))$ on each connected component of $\Omega \setminus Z(\det_2(I-A))$ (so it is branch-independent).

Further, $\log|\det_2(I-A)|$ *is subharmonic on* Ω *and harmonic on* $\Omega\setminus Z(\det_2(I-A))$*; since the (discrete) zero set is polar, it does not affect harmonic-measure boundary trace statements used below. Assume the Carleson energy bound of Lemma 24 for* $\nabla U_{\det_2}$ *on* $Q(I)$*, uniformly up to height* ε_0*. Then the boundary trace* $u_{\det_2}(t) := \lim_{\sigma\downarrow 0} U_{\det_2}(\sigma,t)$ *exists in* $\mathrm{BMO}(I)$ *(hence in* $L^1(I)$*), and in particular*

$$\sup_{0<\sigma\le\varepsilon_0} \|U_{\det_2}(\sigma,\cdot)\|_{L^1(I)} \;<\; \infty.$$

Proof. On $\Omega\setminus Z(\det_2(I-A))$, the function $U_{\det_2} = \log|\det_2(I-A)|$ is harmonic. The Carleson energy hypothesis implies that the measure $|\nabla U_{\det_2}(\sigma,t)|^2\,\sigma\,d\sigma\,dt$ is Carleson on $Q(I)$. By the Fefferman–Stein characterization of BMO boundary traces via Carleson measures for Poisson/harmonic extensions (see, for example, [10, Ch. IV, Thm. 3, p. 159] and [11, Ch. VI, Thm. 3.4]), $U_{\det_2}$ admits nontangential boundary values $U^*_{\det_2} \in \mathrm{BMO}(I)$, hence $U^*_{\det_2} \in L^1(I)$. In particular, $U_{\det_2}(\sigma,\cdot) \to U^*_{\det_2}$ in $L^1(I)$ as $\sigma\downarrow 0$. Moreover, $U_{\det_2}(\sigma,\cdot)$ admits a nontangential boundary trace in $\mathrm{BMO}(I)$ (hence in $L^1(I)$); see [10, 11]. and depends only on the modulus, hence is independent of any choice of analytic branch for $\mathrm{Log}(\det_2(I-A))$. The Carleson energy hypothesis in Lemma 24 gives a Carleson-measure bound for $|\nabla U_{\det_2}|^2\,\sigma\,d\sigma\,dt$ over the Carleson box above I. By the Carleson-measure characterization of BMO boundary traces for harmonic functions on the upper half-plane, this implies that the nontangential boundary trace $u_{\det_2}(t) = \lim_{\sigma\downarrow 0} U_{\det_2}(\sigma,t)$ exists in $\mathrm{BMO}(I)$; in particular $u_{\det_2} \in L^1(I)$. Moreover, the same characterization yields the uniform $L^1(I)$ control $\sup_{0<\sigma\le\varepsilon_0}\|U_{\det_2}(\sigma,\cdot)\|_{L^1(I)} < \infty$. Since the zero set $Z(\det_2(I-A))$ is discrete (hence polar), removing it does not affect harmonic-measure boundary trace statements. □

Lemma 11 (Boundary log-modulus control for ζ on components)**.** *Fix a compact interval* $I \Subset \mathbb{R}$ *and* $\varepsilon_0 \in (0,\frac{1}{2}]$*. Let* U *be a connected component of* $\Omega\setminus Z(\zeta)$ *intersecting* $Q_{\varepsilon_0}(I)$*. Then* ζ *is holomorphic and nonvanishing on* U*, hence* $u(s) = \log|\zeta(s)|$ *is harmonic on* U*. Moreover, the boundary trace* $t \mapsto \log|\zeta(\frac{1}{2}+it)|$ *lies in* $L^1(I)$ *and*

$$\log|\zeta(\tfrac{1}{2}+\varepsilon+it)| \to \log|\zeta(\tfrac{1}{2}+it)| \quad \text{in } L^1(I) \text{ as } \varepsilon\downarrow 0.$$

Proof. Let U be a connected component of $\Omega\setminus Z(\zeta)$ intersecting $Q_{\varepsilon_0}(I)$. Then ζ is holomorphic and nonvanishing on U, hence $u(s) = \log|\zeta(s)|$ is harmonic on U. On the compact strip segment $\{\sigma+it : \sigma\in[\frac{1}{2},\frac{1}{2}+\varepsilon_0],\ t\in I\}$, ζ has only finitely many zeros (counted with multiplicity). For each zero s_k in this compact set, write $\zeta(s) = (s-s_k)^{m_k} g_k(s)$ with g_k holomorphic and nonvanishing in a neighborhood of s_k. Covering the compact strip by finitely many such neighborhoods and a zero-free remainder shows that on the strip

$$\log|\zeta(s)| = \sum_k m_k \log|s-s_k| + O(1),$$

with the $O(1)$ bounded on the strip. For each fixed s_k, the functions $t\mapsto \log|(\frac{1}{2}+\varepsilon+it)-s_k|$ are uniformly $L^1(I)$-bounded for $\varepsilon\in(0,\varepsilon_0]$ and converge in $L^1(I)$ as $\varepsilon\downarrow 0$. Therefore dominated convergence yields the stated $L^1(I)$ convergence $\log|\zeta(\frac{1}{2}+\varepsilon+it)| \to \log|\zeta(\frac{1}{2}+it)|$ as $\varepsilon\downarrow 0$. □

Lemma 12 (Local L^1 control of $\log|F^*|$ on boundary intervals)**.** *Fix a compact interval* $I \Subset \mathbb{R}$ *and* $\varepsilon_0 \in (0,\frac{1}{2}]$*, and set*

$$Q_{\varepsilon_0}(I) := \{\, \tfrac{1}{2}+\sigma+it :\ 0<\sigma\le\varepsilon_0,\ t\in I \,\} \Subset \Omega.$$

Let

$$F(s) := \det_2(I-A(s))\,\frac{s-1}{s\,\zeta(s)}, \qquad s\in\Omega\setminus Z(\zeta).$$

Assume:

(i) $\log|\det_2(I-A(\frac{1}{2}+\varepsilon+it))| \in L^1(I)$ *uniformly for* $\varepsilon\in(0,\varepsilon_0]$*, and the nontangential boundary limit* $\log|\det_2(I-A(\frac{1}{2}+it))|$ *exists in* $L^1(I)$*;*

(ii) *for each connected component U of $\Omega\setminus Z(\zeta)$ intersecting $Q_{\varepsilon_0}(I)$, the function $\log|\zeta(\frac{1}{2}+\varepsilon+it)|$ has an $L^1(I)$-limit as $\varepsilon\downarrow 0$ when restricted to U.*

Then on each such component U, the nontangential boundary values $F^*(t)$ exist for Lebesgue-a.e. $t\in I$, and $\log|F^*(t)|\in L^1_{\mathrm{loc}}(I)$ on U.

Proof. Fix a component U as in the statement. For $s=\frac{1}{2}+\varepsilon+it$ with $0<\varepsilon\le\varepsilon_0$ and $t\in I$, we have

$$\log|F(s)| = \log|\det_2(I-A(s))| + \log|s-1| - \log|s| - \log|\zeta(s)|.$$

Since I is compact and $\varepsilon\in(0,\varepsilon_0]$, the functions $t\mapsto\log|\frac{1}{2}+\varepsilon+it|$ and $t\mapsto\log|-\frac{1}{2}+\varepsilon+it|$ are bounded on I, uniformly in ε; hence $\log|s|$ and $\log|s-1|$ contribute uniformly bounded $L^1(I)$ terms. Assumptions (i)–(ii) therefore imply that $\log|F(\frac{1}{2}+\varepsilon+it)|$ is uniformly in $L^1(I)$ and has an $L^1(I)$ limit as $\varepsilon\downarrow 0$ along U. In particular, after passing to a subsequence if needed, $F(\frac{1}{2}+\varepsilon+it)$ has a nontangential boundary limit for a.e. $t\in I$, and the limiting boundary modulus satisfies $\log|F^*(t)|\in L^1_{\mathrm{loc}}(I)$ on U. □

Lemma 13 (Outer factor from boundary modulus on Ω)**.** *Assume Lemma 7 together with $u\in L^1(\mathbb{R},(1+t^2)^{-1}dt)$. Then there exists a holomorphic function $\mathcal{O}_\zeta$ on Ω, unique up to a unimodular constant, with no zeros on Ω, such that the nontangential boundary values satisfy*

$$ig|\mathcal{O}_\zeta(\tfrac{1}{2}+it)| = ig|F^*(t)| \qquad \textit{for Lebesgue-a.e. } t\in\mathbb{R}.$$

Moreover, $\log|\mathcal{O}_\zeta(s)|$ is the Poisson extension of $u(t)$ from the boundary line $\Re s=\frac{1}{2}$.

Proof. Translate Ω to the right half-plane $\{\,\Re w>0\,\}$ via $w=s-\frac{1}{2}$. Since $u\in L^1(\mathbb{R},(1+t^2)^{-1}dt)$, its Poisson extension $U=\mathcal{P}[u]$ is a harmonic function on Ω with nontangential boundary trace u a.e. Choose a harmonic conjugate V of U on Ω and set $\mathcal{O}_\zeta:=\exp(U+iV)$. Then $\mathcal{O}_\zeta$ is holomorphic and zero-free on Ω, and by Fatou theory its boundary modulus is $e^{u(t)}$ for a.e. t. Uniqueness up to a unimodular constant follows because the ratio of two such outer functions has boundary modulus 1 a.e. and hence is an inner constant; see Garnett [11, Ch. II]. □

Define the outer-normalized ratio

$$\mathcal{J}_{\mathrm{out}}(s) := \frac{F(s)}{\mathcal{O}_\zeta(s)} = \frac{\det_2(I-A(s))}{\mathcal{O}_\zeta(s)\,\zeta(s)}\cdot\frac{s-1}{s}. \tag{4}$$

Then $|\mathcal{J}_{\mathrm{out}}(\frac{1}{2}+it)|=1$ for Lebesgue-a.e. t.

Boundary wedge (P+). Let $w(t):=\operatorname{Arg}\mathcal{J}_{\mathrm{out}}(\frac{1}{2}+it)$ be the boundary phase (defined for a.e. t). We say that (P+) holds if there exists $m\in\mathbb{R}$ such that

$$|w(t)-m|<\frac{\pi}{2} \qquad \text{for Lebesgue-a.e. } t\in\mathbb{R}.$$

Equivalently, $\Re\!\mathfrak{e}g(e^{-im}\mathcal{J}_{\mathrm{out}}(\frac{1}{2}+it))\ge 0$ for Lebesgue-a.e. t.

Remark 14 (Boundary wedge (P+) — not used in the primary proof). If one could establish the global boundary wedge (P+) for $\mathcal{J}_{\mathrm{out}}$, the Schur/Herglotz transport chain below would give an alternative route to Theorem 1. The appendix develops the machinery toward (P+) (phase–velocity identity, CR–Green pairing, Whitney wedge estimates), but a local-to-global step is missing and (P+) as stated would imply zero-freeness for *all* $\Re s>1/2$ (far stronger than the claimed $\Re s\ge 0.6$). The primary proof of Theorem 1 in §4.1 uses a **direct contradiction** that does not invoke (P+).

From a hypothetical (P+) to an interior Schur bound (not used). Fix m witnessing (P+) and set

$$\widetilde{\mathcal{J}}(s) := e^{-im}\mathcal{J}_{\text{out}}(s).$$

Then $|\widetilde{\mathcal{J}}(\frac{1}{2}+it)| = 1$ a.e. and $\Re\,\widetilde{\mathcal{J}}(\frac{1}{2}+it) \geq 0$ a.e. Define the corresponding Cayley field

$$\Theta_{\text{out}}(s) := \frac{2\widetilde{\mathcal{J}}(s)-1}{2\widetilde{\mathcal{J}}(s)+1}.$$

Lemma 15 (Smirnov regularity for $\mathcal{J}_{\text{out}}$ and Θ_{out})**.** *Assume Lemmas 7 and 13. Then $\mathcal{J}_{\text{out}} \in N^+(\Omega\backslash Z(\zeta))$ and admits nontangential boundary values $\mathcal{J}_{\text{out}}(\frac{1}{2}+it)$ for Lebesgue-a.e. t. Consequently, $\widetilde{\mathcal{J}} = e^{-im}\mathcal{J}_{\text{out}} \in N^+(\Omega\setminus Z(\zeta))$, and Θ_{out} admits Lebesgue-a.e. boundary values on $\Re s = \frac{1}{2}$.*

Proof. By Lemma 7, $F \in N^+(\Omega\setminus Z(\zeta))$. By Lemma 13, $\mathcal{O}_\zeta$ is holomorphic, zero-free, and outer on Ω, hence lies in $N^+(\Omega)$. Therefore $\mathcal{J}_{\text{out}} = F/\mathcal{O}_\zeta \in N^+(\Omega\setminus Z(\zeta))$. Multiplication by a unimodular constant preserves N^+ membership, so $\widetilde{\mathcal{J}} \in N^+(\Omega\setminus Z(\zeta))$. The boundary-value statements follow from Smirnov boundary theory (e.g. Garnett [11, Ch. II]). Finally, Θ_{out} is a Möbius transform of $\widetilde{\mathcal{J}}$; its a.e. boundary values exist wherever $2\widetilde{\mathcal{J}}(\frac{1}{2}+it) \neq -1$, which holds a.e. since $\Re\,\widetilde{\mathcal{J}}(\frac{1}{2}+it) \geq 0$ a.e. □

Lemma 16 (Boundary-to-interior Schur transport on components of $\Omega\setminus Z(\zeta)$)**.** *Let U be a connected component of $\Omega\setminus Z(\zeta)$ and let $\Theta \in N^+(U)$ admit nontangential boundary values $\Theta(\frac{1}{2}+it)$ for Lebesgue-a.e. t. If $|\Theta(\frac{1}{2}+it)| \leq 1$ for Lebesgue-a.e. t, then $|\Theta(s)| \leq 1$ for all $s \in U$.*

Proof. For $\Theta \not\equiv 0$, the function $u := \log|\Theta|$ is subharmonic on U and has a harmonic majorant because $\Theta \in N^+(U)$. At Lebesgue points where the nontangential boundary values exist, the boundary hypothesis gives $u(\frac{1}{2}+it) \leq 0$ a.e. Since $Z(\zeta)$ is discrete, it is a polar set, hence has harmonic measure zero for each component U; therefore the Poisson/harmonic-measure domination principle on U yields $u(s) \leq 0$ for all $s \in U$. Thus $|\Theta(s)| \leq 1$ on U. See Garnett [11, Ch. II] and Ransford [8, Ch. 5]. □

Lemma 17 (Boundary-to-interior Herglotz transport on components of $\Omega\setminus Z(\zeta)$)**.** *Let U be a connected component of $\Omega\setminus Z(\zeta)$ and let $G \in N^+(U)$ admit nontangential boundary values $G(\frac{1}{2}+it)$ for Lebesgue-a.e. t. If $\Re\,G(\frac{1}{2}+it) \geq 0$ for Lebesgue-a.e. t, then $\Re\,G(s) \geq 0$ for all $s \in U$.*

Proof. Apply Lemma 16 to the Cayley transform $\Phi(s) := \frac{G(s)-1}{G(s)+1}$ (defined wherever $G \neq -1$) and use continuity to conclude $\Re\,G \geq 0$ on U. Equivalently, one may apply the harmonic-measure domination principle directly to the harmonic function $\Re\,G$ using that $G \in N^+(U)$. □

Proposition 18 (Herglotz/Schur transport)**.** *Assume* (P+) *for $\mathcal{J}_{\text{out}}$ and Lemma 15. Then Θ_{out} is Schur on each connected component of $\Omega\setminus Z(\zeta)$, and*

$$H(s) := \frac{1+\Theta_{\text{out}}(s)}{1-\Theta_{\text{out}}(s)}$$

is Herglotz on $\Omega\setminus Z(\zeta)$ (i.e. $\Re H(s) \geq 0$ there). Moreover, $H(s) = 2\widetilde{\mathcal{J}}(s) = 2e^{-im}\mathcal{J}_{\text{out}}(s)$.

Proof. Let U be any connected component of $\Omega\setminus Z(\zeta)$. By Lemma 15 we have $\widetilde{\mathcal{J}} \in N^+(U)$ and it admits nontangential boundary values on $\Re s = \frac{1}{2}$ for Lebesgue-a.e. t. By (P+) (with the fixed phase m), $\Re\,\widetilde{\mathcal{J}}(\frac{1}{2}+it) \geq 0$ for Lebesgue-a.e. t. Therefore Lemma 17 yields $\Re\,\widetilde{\mathcal{J}}(s) \geq 0$ for all $s \in U$. Since the Cayley map $z \mapsto \frac{2z-1}{2z+1}$ sends the closed right half-plane into the closed unit disc, it follows that $|\Theta_{\text{out}}(s)| \leq 1$ for all $s \in U$. As U was arbitrary, Θ_{out} is Schur on each connected component of $\Omega\setminus Z(\zeta)$. Finally, the Cayley inverse gives

$$H(s) = \frac{1+\Theta_{\text{out}}(s)}{1-\Theta_{\text{out}}(s)} = 2\widetilde{\mathcal{J}}(s) = 2e^{-im}\mathcal{J}_{\text{out}}(s),$$

so H is Herglotz on $\Omega \setminus Z(\zeta)$. □

4.1. **Proof of the main theorem.**

Proof of Theorem 1 (direct contradiction). Fix $\varepsilon > 0$ and suppose for contradiction that $\zeta(\rho_0) = 0$ with $\rho_0 = \beta_0 + i\gamma_0$ and $\beta_0 \geq \frac{1}{2} + \varepsilon$. Set $\delta_0 := \beta_0 - \frac{1}{2} \geq \varepsilon > 0$.

Choice of Whitney parameter (height-dependent). Let $A := Z_0\, C_{\text{test}} \sqrt{2C(\alpha')}$ be the structural constant from the CR–Green and energy bounds (depending only on α' and the window ψ; see (7)). Set $c_\varepsilon := 4\pi/(\varepsilon+1)$, $c_0 := \min\{(c_\varepsilon/(2A))^2,\ 1/2\}$, and

$$c \;:=\; \frac{c_0}{\log\langle\gamma_0\rangle}, \qquad L \;:=\; \min\Big\{\frac{c}{\log\langle\gamma_0\rangle},\ 1\Big\} \;=\; \min\Big\{\frac{c_0}{\log^2\langle\gamma_0\rangle},\ 1\Big\}.$$

Since γ_0 is fixed (by hypothesis), c is a well-defined positive constant. For $|\gamma_0| \geq 2$ one has $\log\langle\gamma_0\rangle \geq 1$, hence $c \leq c_0$ and $L \leq c_0 \leq 1$.

Step 1 (neutralization, sign conventions, and phase-velocity lower bound).
Sign lemma (Blaschke factor phase derivative). For a half-plane Blaschke factor $b(s,\rho) := (s-\rho)/(s-\rho^{\#})$ with $\rho = \frac{1}{2} + \delta + i\gamma$, $\delta > 0$, $\rho^{\#} = 1 - \overline{\rho} = \frac{1}{2} - \delta + i\gamma$, a direct computation gives the boundary phase derivative:

$$-\frac{d}{dt}\operatorname{Arg} b(\tfrac{1}{2} + it, \rho) \;=\; \frac{2\delta}{\delta^2 + (t-\gamma)^2} \;\geq\; 0.$$

(Proof: $b = ((-\delta + i(t-\gamma))/(\delta + i(t-\gamma)))$, so $\operatorname{Arg} b = \pi - 2\arctan((t-\gamma)/\delta)$ and $\frac{d}{dt}\operatorname{Arg} b = -2\delta/(\delta^2 + (t-\gamma)^2) \leq 0$. Hence $-\frac{d}{dt}\operatorname{Arg} b = +2\delta/(\delta^2 + (t-\gamma)^2) \geq 0$. ✓)

Neutralization of the inner reciprocal. Let $D := Q(\alpha'' I)$ be the dilated Whitney box (with $\alpha'' > 2\alpha'$). Let $B_{\text{box}} := \prod_j b(s,\rho_j)^{m_j}$ be the half-plane Blaschke product over the zeros of $\mathcal{I}$ (equivalently, zeros of ζ) **inside the box** D, i.e. those $\rho_j = \beta_j + i\gamma_j$ satisfying **both** $|\gamma_j - \gamma_0| \leq \alpha'' L$ **and** $\delta_j := \beta_j - \frac{1}{2} \leq \alpha'' L$, **with multiplicity** m_j. (The hypothetical zero ρ_0 with $\delta_0 \geq 0.1 \gg \alpha'' L$ does **not** belong to B_{box}.)

Define the **neutralized inner reciprocal**

$$\mathcal{I}_{\text{neut}}(s) \;:=\; \frac{\mathcal{I}(s)}{B_{\text{box}}(s)}\,.$$

Dividing $\mathcal{I}$ by B_{box} **removes** the zeros of $\mathcal{I}$ inside D (each factor $b(s,\rho_j)^{m_j}$ in the denominator cancels the zero at ρ_j). Hence $\mathcal{I}_{\text{neut}}$ is **holomorphic and nonvanishing** on D. Moreover, $|\mathcal{I}_{\text{neut}}| \leq 1$ on Ω (because $\mathcal{I}_{\text{neut}} = \mathcal{I}/B_{\text{box}}$ is a quotient of inner functions and equals a sub-Blaschke product times the singular inner; every factor has modulus ≤ 1). On $\partial\Omega$: $|\mathcal{I}_{\text{neut}}| = |\mathcal{I}|/|B_{\text{box}}| = 1/1 = 1$ a.e.

Set

$$\widetilde{W}(s) \;:=\; -\log|\mathcal{I}_{\text{neut}}(s)| \;\geq\; 0.$$

Then $\widetilde{W}$ is **harmonic** on D, $\widetilde{W} = 0$ on $\sigma = 0$, and $\widetilde{W} = -\log|B_{\text{far}} \cdot S|$ (the same neutralized field as in Proposition 33).

Phase-velocity lower bound (manifestly positive, no pole/zero confusion). Since $\mathcal{I} = e^{i\theta} B_{\mathcal{I}}$ is a pure Blaschke product ($S \equiv 1$ by Proposition 33), and $\mathcal{I}_{\text{neut}} = \mathcal{I}/B_{\text{box}}$ removes only the in-box zeros:

$$-\frac{d}{dt}\operatorname{Arg}\mathcal{I}_{\text{neut}}(\tfrac{1}{2} + it) \;=\; \sum_{\substack{\rho \in Z(\zeta)\cap\Omega \\ \rho \notin D}} m_\rho\, \frac{2\delta_\rho}{\delta_\rho^2 + (t-\gamma_\rho)^2} \;\geq\; 0 \qquad \text{(positive measure).} \tag{5}$$

(Each surviving zero of $\mathcal{I}_{\text{neut}}$ is a **zero**, contributing $+2\delta/(\delta^2 + (t-\gamma)^2) \geq 0$ to $-(\operatorname{Arg}\mathcal{I}_{\text{neut}})'$ by the sign lemma. The in-box zeros have been divided out and do not appear.)

The hypothetical zero ρ_0 has $\delta_0 \geq 0.1 > \alpha'' L$, so $\rho_0 \notin D$ and ρ_0 is **not** divided out. Its Poisson kernel is present in (5):

$$\int_{\mathbb{R}} \psi_{L,\gamma_0}(t)\Big(-\frac{d}{dt}\operatorname{Arg}\mathcal{I}_{\text{neut}}\Big)\,dt \;\geq\; \pi\int_{\gamma_0-L}^{\gamma_0+L}\frac{2\delta_0}{\delta_0^2+(t-\gamma_0)^2}\,dt \;=\; 4\pi\arctan(L/\delta_0) \;\geq\; c_\varepsilon\, L. \tag{6}$$

(Using $\delta_0 \geq \varepsilon$ and $L \leq 1$: $4\pi L/(\delta_0 + L) \geq 4\pi L/(\varepsilon + 1) =: c_\varepsilon\, L > 0$.)

Step 2 (CR–Green upper bound on the neutralized inner reciprocal). Since $\widetilde{W} = -\log|\mathcal{I}_{\text{neut}}|$ is **harmonic** on D and $\widetilde{W} = 0$ on $\sigma = 0$, the Cauchy–Riemann relation gives $\partial_\sigma \widetilde{W}\big|_{\sigma=0} = -\frac{d}{dt}\operatorname{Arg}\mathcal{I}_{\text{neut}}(\frac{1}{2}+it)$ (the same positive measure from (5)). The CR–Green pairing (Proposition 38) applied to $\widetilde{W}$ gives

$$\int_{\mathbb{R}} \psi_{L,\gamma_0}\Big(-\frac{d}{dt}\operatorname{Arg}\mathcal{I}_{\text{neut}}\Big) \;\leq\; Z_0\, C_{\text{test}}\sqrt{E_{\text{neut}}(I)}\cdot L,$$

where

$$E_{\text{neut}}(I) \;:=\; \iint_{Q(\alpha' I)} |\nabla\widetilde{W}|^2\,\sigma\,d\sigma\,dt.$$

(The one-sided inequality is justified because the left side is ≥ 0 by (5).)

By Proposition 33 (boundary bound $M \leq C_* \log\langle\gamma_0\rangle$, interior gradient estimate): By that proposition (with the boundary bound $M \leq C_* \log\langle\gamma_0\rangle$ and the interior gradient estimate):

$$E_{\text{neut}}(I) \;\leq\; C(\alpha')\,\log^2\langle\gamma_0\rangle\,|I|,$$

where $C(\alpha')$ is independent of c (see the "Key independence" remark in Proposition 33). Since $|I| = 2L = 2c_0/\log^2\langle\gamma_0\rangle$:

$$E_{\text{neut}}(I) \;\leq\; C\log^2\langle\gamma_0\rangle\cdot\frac{2c_0}{\log^2\langle\gamma_0\rangle} \;=\; 2Cc_0.$$

Hence

$$Z_0\, C_{\text{test}}\sqrt{E_{\text{neut}}}\cdot L \;\leq\; A\sqrt{c_0}\;\cdot\; L, \tag{7}$$

where $A := Z_0\, C_{\text{test}}\sqrt{2C(\alpha')}$ is independent of c_0 and γ_0.

Step 3 (contradiction). Combining (6) and (7): $c_\varepsilon\, L \leq A\sqrt{c_0}\, L$, hence $c_\varepsilon \leq A\sqrt{c_0}$. Choosing $c_0 := (c_\varepsilon/(2A))^2$ gives $A\sqrt{c_0} = c_\varepsilon/2 < c_\varepsilon$. **Contradiction.** (Here $c_\varepsilon = 4\pi/(\varepsilon+1) > 0$ depends only on ε, and $c_0 = (c_\varepsilon/(2A))^2$ depends only on ε and the structural constants A, α'.)

Small-height case ($|\gamma_0| \leq 2$): vacuous. The first nontrivial zero of ζ has $|\gamma| \approx 14.13$ (classical; see Titchmarsh [3, Ch. X]). Hence there are **no** nontrivial zeros with $|\gamma_0| \leq 2$, and the contradiction hypothesis is vacuously false in this range. No computation is required. □

Remark 19 (Alternative route via the boundary wedge (P+)). The same conclusion follows (with the same inputs) by the boundary-wedge route of §4–Appendix A: Theorem 14 ⇒ Proposition 18 ⇒ Corollary 6. The direct-contradiction proof above is logically simpler because it avoids the global boundary-wedge statement (P+), the Smirnov boundary transport, and the interior Herglotz/Schur mechanism; it needs only the *localized* phase-velocity lower bound from a single hypothetical zero.

TABLE 1. Supplementary computational artifacts (cross-checks; not logically required for the proof).

Artifact	**Parameter**	**Value**
Rectangle certification (`theta_certify`)		
Domain	$[\sigma_{\min}, \sigma_{\max}] \times [t_{\min}, t_{\max}]$	$[0.6, 0.7] \times [0, 20]$
upper bound	$\max \lvert\Theta_{\text{proj}}\rvert$	0.9999928763
Safety margin	$1 - \theta_{\text{hi}}$	7.12×10^{-6}
Status	`ok`	`true`
Boxes processed		380,764
Precision	(bits)	260
Gauge		`outer_zeta_proj`
Pick certificate (`pick_certify`, $\sigma_0 = 0.599$)		
Matrix size	N	16
Spectral separation	δ_{cert}	0.594
SPD at origin	$P_N \succ 0$	`true`
Coefficient count	N_{coeff}	128
Tail sum (diagnostic)	$\sum_{16}^{127} \lvert a_n\rvert$	0.67
Gauge		`raw_zeta`
Pick certificate (`pick_certify`, $\sigma_0 = 0.6$)		
Matrix size	N	16
Spectral separation	δ_{cert}	0.594
SPD at origin	$P_N \succ 0$	`true`
Coefficient count	N_{coeff}	128
Tail sum (diagnostic)	$\sum_{16}^{127} \lvert a_n\rvert$	0.67
Gauge		`raw_zeta`
Pick certificate (`pick_certify`, $\sigma_0 = 0.7$)		
Matrix size	N	16
Spectral separation	δ_{cert}	0.627
SPD at origin	$P_N \succ 0$	`true`
Coefficient count	N_{coeff}	128
Tail sum (diagnostic)	$\sum_{16}^{127} \lvert a_n\rvert$	0.61
Gauge		`raw_zeta`

CONCLUSION AND LIMITATIONS (UNCONDITIONAL STATUS)

We prove the Riemann Hypothesis: $\zeta(s) \neq 0$ for $\Re s > 1/2$ (Theorem 1).

The argument is analytic and self-contained. Zeros of ζ in Ω are converted into zeros of the analytic inner reciprocal $\mathcal{I} := B^2/\mathcal{J}_{\text{out}}$ (Lemma 31), whose non-circular boundedness $|\mathcal{I}| \leq 1$ follows from the Phragmén–Lindelöf principle (no assumption about ζ-zeros is used). The resulting nonnegative potential $W = -\log|\mathcal{I}| \geq 0$ provides a Whitney-box energy bound (Proposition 33) with growth $E(I) \leq C \log^2\langle t_0\rangle\, |I|$.

Unconditional status. The singular inner factor S of the inner reciprocal $\mathcal{I}$ is proved trivial ($S \equiv 1$) in Proposition 33, Step 2, using only the polynomial growth of ζ on Ω (convexity bound) and the standard N^+ criterion (Garnett [11, Ch. II]). With $S \equiv 1$, the neutralized boundary bound is $M = O(\log\langle t_0\rangle)$ with constant independent of c, and the height-dependent Whitney parameter $c = c_0/\log\langle\gamma_0\rangle$ collapses the $\log^2$ factor to a constant, yielding the unconditional contradiction $11 \leq A\sqrt{c_0} = 11/\sqrt{2} < 11$.

Computer assistance. The proof is **purely analytic**; no computation is logically required. (The small-height case $|\gamma_0| \leq 2$ is vacuous because the first nontrivial zero has $|\gamma| \approx 14.13$.) The shipped

ball-arithmetic certificates (Table 1) are included as supplementary cross-checks for transparency and reproducibility.
Limitations and scope. We do not claim the Riemann Hypothesis here.
Scope. The theorem establishes $\zeta(s) \neq 0$ for every $\Re s > 1/2$, which is equivalent to the Riemann Hypothesis for the open half-plane. The critical line $\Re s = 1/2$ itself is not covered (zeros on the critical line are known to exist and are not excluded by this method).

Statements and Declarations

Competing interests. The author declares no competing interests.
Data and materials availability. For reproducibility, the verification is with respect to the shipped bundle (and its SHA-256 manifest, if provided). If any file is updated after theory revision, the verification should be rerun and the manifest regenerated. All computational artifacts used for supplementary cross-checks are included in the handoff bundle (and mirrored in the repository):

```
compute/artifacts/theta_certify_sigma06_07_t0_20_outer_zeta_proj.json
compute/artifacts/pick_sigma0599_raw_zeta_N16.json
compute/artifacts/pick_sigma06_raw_zeta_N16.json
compute/artifacts/pick_sigma07_raw_zeta_N16.json
compute/verify_attachment_arb.py
```

Remark 20. The verifier is based on rigorous ball arithmetic (ARB via `python-flint`) and is intended to be independently verifiable. Appendix B records the verification manifest and a minimal command sequence for running the verifier on a fresh machine.

Appendix A. Proof of the boundary wedge certificate (P+)

This appendix develops the analytic machinery (phase–velocity identity, CR–Green pairing, Whitney-box energy estimates) used in the direct-contradiction proof of Theorem 1 (§4.1). The boundary wedge (P+) discussed in Remark 14 is **not** established here; the primary proof uses a localized contradiction that does not require the global (P+) statement.

A.1. **Statement, standing notation, and domains.** This subsection fixes the ambient domain, boundary conventions, Whitney geometry, and the meaning of boundary limits, so later phase and energy identities are unambiguous.

Throughout Appendix A we work in the right half-plane

$$\Omega := \{s \in \mathbb{C} : \Re s > \tfrac{1}{2}\},$$

with boundary line $\partial\Omega = \{\frac{1}{2} + it : t \in \mathbb{R}\}$. All analytic objects are understood componentwise on $\Omega \setminus Z$, where Z denotes the relevant zero/pole set, so that branches of log and Arg are well-defined on each connected component.

For a compact interval $I \subset \mathbb{R}$ and a dilation parameter $\alpha > 1$ we write $Q_\alpha(I)$ for the Whitney box based on I, and we use the weighted area measure $\sigma\, dt\, d\sigma$ on Ω, where $\sigma := \Re s - \frac{1}{2}$.

The machinery below supports the direct-contradiction proof of Theorem 1; the global boundary wedge (P+) of Remark 14 is **not** established here. The proof proceeds by: (i) a quantitative phase–velocity identity for the boundary phase of $\mathcal{J}_{\text{out}}$, (ii) a Green/Cauchy–Riemann pairing on Whitney boxes, (iii) a Carleson-type energy bound for a logarithmic derivative, and (iv) a quantitative wedge criterion converting windowed phase control into a.e. wedge inclusion.

In the phase–velocity identity below, we will show that $-w'$ is positive (in the sense of distributions): a locally finite measure plus a discrete atomic part encoding the off–critical zero data that enter the boundary phase derivative.

A.2. **A quantitative wedge criterion from Whitney-local control.** We state the wedge target (P+) in a form suited to local Whitney-box estimates and record the boundary conventions used throughout Appendix A.

We work on the boundary line $\Re s = \frac{1}{2}$ and use the following conventions.

- *Wedge.* For an aperture parameter $\alpha \in (0, \frac{\pi}{2})$ and a center angle $m \in \mathbb{R}$, write
$$W_{m,\alpha} := \{z \in \mathbb{C} : \ |\operatorname{Arg}(e^{-im}z)| \le \alpha\}.$$
Thus (P+) is the Lebesgue-a.e. inclusion $\mathcal{J}_{\text{out}}(\frac{1}{2} + it) \in W_{m,\alpha}$ for some fixed $\alpha < \frac{\pi}{2}$ and some $m \in \mathbb{R}$.
- *Whitney / Carleson boxes.* For an interval $I \subset \mathbb{R}$, write the Carleson box $S(I) := \{\frac{1}{2} + \sigma + it : 0 < \sigma \le |I|, \ t \in I\}$. A Whitney box means a box of comparable width and height, e.g. $\{\frac{1}{2} + \sigma + it : \ \sigma \in [a|I|, b|I|], \ t \in I\}$ with fixed $0 < a < b$.
- *Meaning of "a.e."* Unless explicitly stated otherwise, "a.e." refers to Lebesgue measure dt on $\mathbb{R}$.

Lemma 21 (Outer normalizer from boundary log-modulus)**.** *Let $u \in L^1(\mathbb{R}, (1+t^2)^{-1}dt)$ be real-valued. Then there exists an outer function O on Ω (zero-free and holomorphic on Ω) whose nontangential boundary values satisfy*
$$|O(\tfrac{1}{2} + it)| = e^{u(t)} \quad \textit{for a.e. } t \in \mathbb{R}.$$
Moreover O is unique up to a unimodular constant.

Proof. Define the Poisson extension U of u to Ω by
$$U(\tfrac{1}{2} + \sigma + it) \ := \ \frac{1}{\pi} \int_{\mathbb{R}} u(\tau) \, \frac{\sigma}{\sigma^2 + (t-\tau)^2} \, d\tau, \qquad \sigma > 0.$$
The weighted integrability $u \in L^1(\mathbb{R}, (1+t^2)^{-1}dt)$ ensures the integral converges and that U is harmonic on Ω. Let V be a harmonic conjugate of U on Ω (defined up to an additive constant), and set
$$O(s) \ := \ \exp ig(U(s) + iV(s)).$$
Then O is holomorphic and zero-free on Ω. By the nontangential boundary limit theorem for Poisson extensions of L^1_{loc} boundary data, one has $U(\frac{1}{2} + \varepsilon + it) \to u(t)$ for a.e. t as $\varepsilon \downarrow 0$; hence the nontangential boundary values satisfy $|O(\frac{1}{2} + it)| = e^{u(t)}$ for a.e. t; see Duren [9, Ch. II] or Garnett [11, Ch. II]. Uniqueness up to unimodular constant follows because the ratio of two such outer functions has a.e. boundary modulus 1 and hence is an inner constant. (Source.) This is the outer-function construction in half-planes; see Duren *H^p Spaces*, Ch. II, or Garnett *Bounded Analytic Functions*, Ch. II. □

A.3. **Phase–velocity identity (quantitative form) and boundary passage.** We establish the boundary phase–velocity relation for the outer-normalized ratio, and record the precise sense in which derivatives and boundary traces are taken.

Lemma 22 (Outer–Hilbert boundary identity)**.** *Let $u \in L^1_{\text{loc}}(\mathbb{R})$ and let O be an outer function on Ω whose boundary modulus satisfies $|O(\frac{1}{2} + it)| = e^{u(t)}$ for a.e. t. Let $w(t) := \operatorname{Arg} O(\frac{1}{2} + it)$ denote the boundary argument (defined modulo an additive constant). Then, in $\mathcal{D}'(\mathbb{R})$,*
$$\frac{d}{dt} w(t) = \mathcal{H}[u'](t),$$
where $\mathcal{H}$ is the boundary Hilbert transform on $\mathbb{R}$ (as a continuous operator $\mathcal{D}'(\mathbb{R}) \to \mathcal{D}'(\mathbb{R})$) and u' is the distributional derivative.

Proof. Write $\log O = U + iV$ on Ω, where $U = \Re \log O$ is harmonic and $V = \Im \log O$ is its harmonic conjugate (on each component of $\Omega \setminus Z(O)$, fixing a branch of log). The boundary trace satisfies $U(\frac{1}{2} + \cdot) = u$ in $\mathcal{D}'(\mathbb{R})$, and the conjugate boundary trace is $V(\frac{1}{2} + \cdot) = \mathcal{H}[u]$ in $\mathcal{D}'(\mathbb{R})$ (up to an additive constant). Differentiating in t in the sense of distributions gives

$$\frac{d}{dt} \operatorname{Arg} O(\tfrac{1}{2} + it) = \partial_t V(\tfrac{1}{2} + it) = \mathcal{H}[\partial_t u](t) = \mathcal{H}[u'](t),$$

since differentiation commutes with $\mathcal{H}$ on $\mathcal{D}'(\mathbb{R})$. □

Lemma 23 (Smoothed distributional bound for $\partial_\sigma \Re \log \det_2$)**.** *Let $I \Subset \mathbb{R}$ be a compact interval and fix $\varepsilon_0 \in (0, \frac{1}{2}]$. There exists a finite constant*

$$C_* \;:=\; \sum_p \sum_{k \geq 2} \frac{p^{-k/2}}{k^2 \, \log p} \;<\; \infty$$

such that for all $\sigma \in (\frac{1}{2}, \frac{1}{2} + \varepsilon_0]$ and every $\varphi \in C_c^2(I)$,

$$\Big| \int_{\mathbb{R}} \varphi(t) \, \partial_\sigma \Re \log \det_2 \big(I - A(\sigma + it)\big) \, dt \Big| \;\leq\; C_* \, \|\varphi''\|_{L^1(I)}.$$

Proof. For $\sigma > \frac{1}{2}$ one has the absolutely convergent expansion

$$\partial_\sigma \, \Re \log \det_2 \big(I - A(\sigma + it)\big) \;=\; \sum_p \sum_{k \geq 2} (\log p) \, p^{-k\sigma} \cos(kt \log p).$$

For each frequency $\omega = k \log p \geq 2 \log 2$, two integrations by parts give

$$\Big| \int_{\mathbb{R}} \varphi(t) \cos(\omega t) \, dt \Big| \;\leq\; \frac{\|\varphi''\|_{L^1(I)}}{\omega^2}.$$

Summing the resulting majorant yields

$$\Big| \int \varphi \, \partial_\sigma \Re \log \det_2 \, dt \Big| \;\leq\; \|\varphi''\|_{L^1} \sum_p \sum_{k \geq 2} \frac{(\log p) \, p^{-k\sigma}}{(k \log p)^2} \;\leq\; \|\varphi''\|_{L^1} \sum_p \sum_{k \geq 2} \frac{p^{-k/2}}{k^2 \, \log p},$$

uniformly for $\sigma \in (\frac{1}{2}, \frac{1}{2} + \varepsilon_0]$, since the rightmost double series converges. □

Lemma 24 (Arithmetic Carleson energy)**.** *Let*

$$U_{\det_2}(\sigma, t) \;:=\; \Re \log \det_2 \Big(I - A(\tfrac{1}{2} + \sigma + it)\Big) \;=\; -\sum_p \sum_{k \geq 2} \frac{p^{-k/2}}{k} \, e^{-k \log p \, \sigma} \, \cos(k \log p \, t), \qquad \sigma > 0,$$

where the series converges absolutely for every $\sigma > 0$. Then for every interval $I \subset \mathbb{R}$ with Carleson box $Q(I) := I \times (0, |I|]$,

$$\iint_{Q(I)} |\nabla U_{\det_2}|^2 \, \sigma \, dt \, d\sigma \;\leq\; \frac{|I|}{4} \sum_p \sum_{k \geq 2} \frac{p^{-k}}{k^2} \;=:\; K_0 \, |I|, \qquad K_0 := \frac{1}{4} \sum_p \sum_{k \geq 2} \frac{p^{-k}}{k^2} < \infty.$$

Proof. For a single mode $b \, e^{-\omega\sigma} \cos(\omega t)$ one has $|\nabla|^2 = b^2 \omega^2 e^{-2\omega\sigma}$, hence

$$\int_0^{|I|} \int_I |\nabla|^2 \, \sigma \, dt \, d\sigma \;\leq\; |I| \cdot \sup_{\omega > 0} \int_0^{|I|} \sigma \, \omega^2 e^{-2\omega\sigma} d\sigma \cdot b^2 \;\leq\; \tfrac{1}{4} \, |I| \, b^2.$$

With $b = p^{-k/2}/k$ and $\omega = k \log p$, summing over (p, k) gives the claim and the finiteness of K_0. □

Whitney scale and short–interval zero counts. Throughout the boundary-certificate route we work on Whitney boxes based at height T with

$$L = L(T) := \min\Big\{\frac{c}{\log\langle T\rangle},\ L_\star\Big\}, \qquad \langle T\rangle := \sqrt{1+T^2}, \qquad c \in (0,1] \text{ fixed.}$$

The only input about the *number* of zeros used below is the classical consequence of Riemann–von Mangoldt:

$$N(T;H) \ := \ \#\{\rho = eta + i\gamma :\ \gamma \in [T, T+H]\} \ \le\ C_{\mathrm{RvM}}\,(1+H)\,\log\langle T\rangle, \tag{8}$$

for all $T \ge 2$ and $H > 0$, where C_{RvM} is an absolute constant. (This follows from $N(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + O(\log T)$ and the $O(\log T)$ error in the Riemann–von Mangoldt formula; see [3].) On Whitney scale $H = 2L = 2c/\log\langle T\rangle$ the bound gives $N(T;2L) \le C_{\mathrm{RvM}}(1+2c)\log\langle T\rangle = O(\log\langle T\rangle)$, not $O(1)$.

Lemma 25 (Annular Poisson–balayage L^2 bound)**.** *Let $I = [T-L, T+L]$, $Q_\alpha(I) = I \times (0, \alpha L]$, and fix $k \ge 1$. For $\mathcal{A}_k := \{\rho = eta + i\gamma :\ 2^k L < |T-\gamma| \le 2^{k+1}L\}$ set*

$$V_k(\sigma,t) := \sum_{\rho\in\mathcal{A}_k} \frac{\sigma}{(t-\gamma)^2+\sigma^2}.$$

Then

$$\iint_{Q_\alpha(I)} V_k(\sigma,t)^2\,\sigma\,dt\,d\sigma \ \ll_\alpha\ |I|\,4^{-k}\,\nu_k,$$

where $\nu_k := \#\mathcal{A}_k$, and the implicit constant depends only on α.

Proof. Write $K_\sigma(x) := \sigma/(x^2+\sigma^2)$ and $V_k = \sum_{\rho\in\mathcal{A}_k} K_\sigma(\cdot-\gamma)$. Integrate over $t \in I$ first. For the diagonal terms, using $|t-\gamma| \ge 2^k L - L \ge 2^{k-1}L$ for $t \in I$ and $k \ge 1$,

$$\int_I K_\sigma(t-\gamma)^2\,dt = \sigma^2\int_I \frac{dt}{ig((t-\gamma)^2+\sigma^2 ig)^2} \ \le\ \frac{L}{(2^{k-1}L)^2}\,\sigma.$$

Multiplying by the area weight σ and integrating $\sigma \in (0,\alpha L]$ gives a contribution $\ll_\alpha |I|\,4^{-k}$ per γ, hence $\ll_\alpha |I|\,4^{-k}\nu_k$ after summing. For off-diagonal terms, for $i \ne j$ one has on I that $K_\sigma(t-\gamma_j) \le \sigma/(2^{k-1}L)^2$, hence

$$\int_I K_\sigma(t-\gamma_i)K_\sigma(t-\gamma_j)\,dt \ \le\ \frac{\sigma}{(2^{k-1}L)^2}\int_{\mathbb{R}} K_\sigma(t-\gamma_i)\,dt = \frac{\pi\sigma}{(2^{k-1}L)^2},$$

and integrating $\sigma \in (0,\alpha L]$ with the extra factor σ yields $\ll_\alpha |I|\,4^{-k}$. Summing over pairs (i,j) via a Schur test gives the stated bound (absorbing constants into $\ll_\alpha$). □

A.4. **Quantitative phase–velocity identity.** This subsection derives the quantitative identity linking the distribution $-w'$ to the off-axis zero data, in a form usable under Whitney localization.

Lemma 26 (Distributional phase–velocity identity for outer data)**.** *Let $U^* \in L^1_{\mathrm{loc}}(\mathbb{R})$ and define $w := \mathcal{H}[U^*] \in \mathcal{D}'(\mathbb{R})$ by the duality*

$$-\langle w, \varphi'\rangle \ = \ \int_{\mathbb{R}} U^*(t)\,(\mathcal{H}\varphi)'(t)\,dt \qquad \forall\,\varphi \in C_c^\infty(\mathbb{R}).$$

Then $w' = \mathcal{H}[(U^*)']$ in $\mathcal{D}'(\mathbb{R})$.

Proof. Let η_ε be a mollifier and set $U^*_\varepsilon := U^* * \eta_\varepsilon \in C^\infty(\mathbb{R})$. For smooth data one has $(\mathcal{H}[U^*_\varepsilon])' = \mathcal{H}[(U^*_\varepsilon)']$ pointwise. Since $U^*_\varepsilon \to U^*$ in L^1_{loc}, we have $(U^*_\varepsilon)' \to (U^*)'$ in $\mathcal{D}'$, and the Hilbert transform is continuous on $\mathcal{D}'$. Passing to the limit yields $w' = \mathcal{H}[(U^*)']$ in $\mathcal{D}'$. □

Lemma 27 (Local L^1 control for $\log|\xi|$ along vertical approach)**.** *Fix a compact interval $I \Subset \mathbb{R}$. Then the family $t \mapsto \log|\xi(\frac12+\varepsilon+it)|$ is bounded in $L^1(I)$ uniformly for $\varepsilon \in (0,1]$. Moreover, for $\varepsilon, \varepsilon' \downarrow 0$ the difference $\log|\xi(\frac12+\varepsilon+it)| - \log|\xi(\frac12+\varepsilon'+it)|$ tends to 0 in $L^1(I)$.*

Proof. Write ξ in Hadamard form $\xi(s) = e^{a+bs}\prod_\rho igl(1-\frac{s}{\rho}igr)e^{s/\rho}$, where the product runs over nontrivial zeros ρ of ζ. Fix $I = [T_0, T_1] \Subset \mathbb{R}$ and $\varepsilon \in (0,1]$. Split the zeros into a finite set $\mathcal{Z}_R := \{\rho : \ |\Im\rho| \le R\}$ and the complement, with $R \ge 2 + \max(|T_0|, |T_1|)$. For $\rho \in \mathcal{Z}_R$, the map $t \mapsto \log|(\frac{1}{2}+\varepsilon+it)-\rho|$ lies in $L^1(I)$, with an $L^1(I)$ bound depending only on I and $\mathcal{Z}_R$ (local integrability of $\log|t-\gamma|$ near $\gamma = \Im\rho$). For $\rho \notin \mathcal{Z}_R$ and $t \in I$, one has $|(\frac{1}{2}+\varepsilon+it)/\rho| \ll_I 1/|\rho|$, so

$$\log\Big|\Big(1-\frac{\frac{1}{2}+\varepsilon+it}{\rho}\Big)e^{(\frac{1}{2}+\varepsilon+it)/\rho}\Big| = O_I igl(|\rho|^{-2}igr),$$

uniformly in $t \in I$ and $\varepsilon \in (0,1]$. Since $\sum_\rho |\rho|^{-2} < \infty$ (order 1 entire function), the tail contributes an absolutely convergent $L^\infty(I)$ error uniformly in ε. Combining these bounds gives $\sup_{\varepsilon\in(0,1]} \|\log|\xi(\frac{1}{2}+\varepsilon+i\cdot)|\|_{L^1(I)} < \infty$.

For the Cauchy property, write the difference as a sum over the same factorization. The finite set $\mathcal{Z}_R$ contributes a term that tends to 0 in $L^1(I)$ as $\varepsilon, \varepsilon' \downarrow 0$ by dominated convergence away from the finitely many points $t = \Im\rho$ and the local integrability of $\log|t-\Im\rho|$. The tail is uniformly $O_I\Big(\sum_{\rho\notin\mathcal{Z}_R}|\rho|^{-2}\Big)$ and hence uniformly small; letting $R \to \infty$ yields the $L^1(I)$-Cauchy claim. □

Lemma 28 (Local bounded-type control for $\mathcal{J}$)**.** *Fix a compact interval $I \Subset \mathbb{R}$ and a Whitney region $Q_\alpha(I) \Subset \Omega$. Assume that $\det_2(I - A(s))$ is holomorphic and nonvanishing on a neighborhood of $Q_\alpha(I)$, and that the arithmetic Carleson energy bound of Lemma 24 holds on $Q_\alpha(I)$. Then the ratio $\mathcal{J} = \det_2(I-A)/(\mathcal{O}\,\xi)$ constructed in Theorem 29 belongs to the Nevanlinna class (bounded type) on $Q_\alpha(I)$.*

Proof. Let $D := Q_\alpha(I)$ and write $\sigma = \Re s - \frac{1}{2}$. On D (simply connected), the hypotheses ensure that $\det_2(I-A)$ is holomorphic and nonvanishing, so $U_{\det_2} := \Re\log\det_2(I-A)$ is harmonic on D. The Carleson energy bound of Lemma 24 implies that $|\nabla U_{\det_2}|^2\,\sigma\,dt\,d\sigma$ is a Carleson measure on D. By the Fefferman–Stein/Carleson characterization of BMO boundary traces for harmonic functions on the half-plane (see Stein [10, Ch. IV] or Garnett [11, Ch. VI]), $U_{\det_2}$ admits nontangential boundary traces in $\mathrm{BMO}(I) \subset L^1(I)$. Hence $\det_2(I-A)$ is of bounded type on D.

Next, ξ is entire, so it is holomorphic on a neighborhood of $\overline{D}$ and therefore bounded on D; in particular $\xi \in H^\infty(D)$ and is of bounded type on D (and $1/\xi$ is meromorphic of bounded type on D).

Finally, the normalizing outer function $\mathcal{O}$ from Theorem 29 is holomorphic and zero-free on Ω, hence of bounded type on D. Since the Nevanlinna class is closed under products and quotients (where defined), it follows that $\mathcal{J} = \det_2(I-A)/(\mathcal{O}\,\xi)$ is of bounded type on D. □

Theorem 29 (Quantified phase–velocity identity and boundary passage)**.** *Let*

$$u_\varepsilon(t) := \log|\det_2(I - A(\tfrac{1}{2}+\varepsilon+it))| - \log|\xi(\tfrac{1}{2}+\varepsilon+it)|.$$

Then u_ε is uniformly L^1-bounded and Cauchy on every compact $I \Subset \mathbb{R}$ as $\varepsilon \downarrow 0$, hence $u_\varepsilon \to u$ in $L^1_{\mathrm{loc}}(\mathbb{R})$. Let $\mathcal{O}$ be the outer function on Ω with boundary modulus e^u and normalization $\mathcal{O}(\frac{3}{2}) > 0$, and set

$$\mathcal{J}(s) := \frac{\det_2(I - A(s))}{\mathcal{O}(s)\,\xi(s)}.$$

Then $|\mathcal{J}(\frac{1}{2}+it)| = 1$ for a.e. $t \in \mathbb{R}$. By Lemma 28, $\mathcal{J}$ is of bounded type on every Whitney region $Q_\alpha(I)$. Let $w \in \mathcal{D}'(\mathbb{R})$ denote the distributional boundary phase of $\mathcal{J}$ (defined modulo an additive constant). Then, for every compact interval $I \Subset \mathbb{R}$ and every nonnegative $\phi \in C_c^\infty(I)$,

$$\int_I \phi(t)\,(-w'(t))\,dt \;=\; \pi\int_I \phi(t)\,d\mu_{\mathrm{off}}(t) \;+\; \pi\int_I \phi(t)\,d\nu_{\mathrm{sing}}(t) \;+\; \pi\sum_{\gamma\in I} m_\gamma\,\phi(\gamma), \tag{9}$$

where:

- *μ_{off} is the Poisson balayage of the off–critical zeros $\rho = eta + i\gamma$ of ζ with $eta > \frac{1}{2}$, counted with multiplicity m_ρ;*
- *ν_{sing} is the (possibly zero) singular boundary measure of any singular inner factor in the canonical factorization of $\mathcal{J}$ on Ω; and*
- *the discrete sum ranges over boundary zeros/poles on $\Re s = \frac{1}{2}$, written $s = \frac{1}{2} + i\gamma$, with multiplicities m_γ.*

Proof. The L^1_{loc} convergence $u_\varepsilon \to u$ is as stated. The outer function $\mathcal{O}$ exists by Lemma 21.

Under the bounded-type hypothesis, $\mathcal{J}$ admits the canonical half-plane factorization into a unimodular constant, a Blaschke product over zeros in Ω, a (possibly trivial) singular inner factor, and an outer factor. Since $|\mathcal{J}(\frac{1}{2} + it)| = 1$ a.e., the outer factor is unimodular constant. Taking the distributional boundary argument w and differentiating in $\mathcal{D}'$, each factor contributes additively: the Blaschke product yields the Poisson balayage measure μ_{off}, the singular inner factor yields ν_{sing}, and boundary zeros/poles yield atomic Dirac masses. This is the phase-derivative computation for bounded-type functions on a half-plane; see, e.g., Garnett *Bounded Analytic Functions*, Ch. II, or Koosis *The Logarithmic Integral*, Vol. I. □

A.5. **Final assembly ingredients for the boundary wedge certificate (P+).** This subsection collects the three final analytic ingredients—Poisson plateau, Whitney box pairing, and Carleson-energy control—used to conclude (P+).

Poisson plateau lower bound. We prove the key lower bound for the Poisson kernel averaged over admissible windows, which converts discrete off-axis contributions into a uniform wedge aperture.

Lemma 30 (Poisson plateau lower bound)**.** *Let $\psi \in C_c^\infty(\mathbb{R})$ be even with $\psi \equiv 1$ on $[-1, 1]$ and* $\operatorname{supp}\psi \subset [-2, 2]$. *Then*

$$c_0(\psi) \;:=\; \inf_{0<b\le 1,\ |x|\le 1} (P_b * \psi)(x) \;\ge\; \frac{1}{2\pi}\arctan 2 \;>\; 0.$$

Proof. Since $\psi \ge \mathbf{1}_{[-1,1]}$, it suffices to compute $(P_b * \mathbf{1}_{[-1,1]})(x)$. For $|x| \le 1$,

$$(P_b * \mathbf{1}_{[-1,1]})(x) = \frac{1}{\pi}\int_{-1}^{1} \frac{b}{b^2 + (x-y)^2}\,dy = \frac{1}{2\pi}\Big(\arctan\frac{1-x}{b} + \arctan\frac{1+x}{b}\Big).$$

This expression is minimized over $0 < b \le 1$, $|x| \le 1$, at $(x, b) = (1, 1)$, yielding $\frac{1}{2\pi}\arctan 2$. □

From phase–velocity and CR–Green to (P+). We combine the phase–velocity identity with a Cauchy–Riemann/Green pairing on Whitney boxes to obtain the windowed phase control that underlies (P+).

Carleson energy bound for the logarithmic derivative. We now pass to the load-bearing box-energy control used in the wedge assembly. The key device is the *analytic inner reciprocal* $\mathcal{I} := B^2/\mathcal{J}_{\text{out}}$, which converts the poles of $\mathcal{J}_{\text{out}}$ (at ζ-zeros) into harmless zeros, yielding an honest inner function and a non-circular source of positivity for the Whitney gradient estimate.

Lemma 31 (Inner reciprocal and nonnegative potential)**.** *Let $\mathcal{J}_{\text{out}}$ be as in* (4) *and $B(s) := (s-1)/s$. Define*

$$\mathcal{I}(s) \;:=\; \frac{B(s)^2}{\mathcal{J}_{\text{out}}(s)} \;=\; \frac{B(s)\,\mathcal{O}_\zeta(s)\,\zeta(s)}{\det_2(I - A(s))}\,.$$

Then:

(1) *$\mathcal{I}$ is **holomorphic** on Ω. (The simple pole of ζ at $s = 1$ is canceled by B; zeros of ζ become zeros of $\mathcal{I}$; the denominator $\det_2(I - A)$ is nonvanishing on Ω.)*
(2) *$|\mathcal{I}(\frac{1}{2} + it)| = 1$ for Lebesgue-a.e. t. (On $\partial\Omega$: $|B| = 1$ and $|\mathcal{J}_{\text{out}}| = 1$ a.e.)*
(3) *$|\mathcal{I}(s)| \le 1$ for all $s \in \Omega$. (Phragmén–Lindelöf: $\log|\mathcal{I}|$ is subharmonic on Ω with boundary trace 0 a.e. and at most polynomial growth; see below.)*

In particular, the function

$$W(s) \;:=\; -\log|\mathcal{I}(s)| \;\geq\; 0 \qquad (s\in\Omega)$$

is nonnegative, and one has the identity

$$U(s) \;:=\; \log|\mathcal{J}_{\mathrm{out}}(s)| \;=\; 2\log|B(s)| \;+\; W(s) \qquad (s\in\Omega\setminus Z(\zeta)).$$

Proof. Part (1). Write $\mathcal{I} = B\,\mathcal{O}_\zeta\,\zeta/\det_2(I-A)$. The factor $B\zeta = (s-1)\zeta(s)/s$ is holomorphic on Ω (the simple pole of ζ at $s=1$ is canceled by the zero of $s-1$, and $s=0\notin\Omega$). The remaining factors $\mathcal{O}_\zeta$ (outer, zero-free) and $1/\det_2(I-A)$ (nonvanishing by (1)) are holomorphic on Ω. Hence $\mathcal{I}$ is holomorphic on Ω, with zeros exactly at the nontrivial zeros of ζ in Ω (same multiplicities).

Part (2). On $\partial\Omega$: $|B(\frac12+it)|^2 = |(-\frac12+it)/(\frac12+it)|^2 = (\frac14+t^2)/(\frac14+t^2) = 1$, and $|\mathcal{J}_{\mathrm{out}}(\frac12+it)| = 1$ a.e. by construction. Hence $|\mathcal{I}(\frac12+it)| = |B|^2/|\mathcal{J}_{\mathrm{out}}| = 1$ a.e.

Part (3): $|\mathcal{I}|\leq 1$ via Phragmén–Lindelöf. Since $\mathcal{I}$ is holomorphic on Ω, $u := \log|\mathcal{I}|$ is subharmonic on Ω.

Boundary trace. For $\varepsilon>0$ set $s_\varepsilon := \frac12+\varepsilon+it$. Each factor of $\mathcal{I} = B\,\mathcal{O}_\zeta\,\zeta/\det_2(I-A)$ has L^1_{loc}-convergent log-modulus as $\varepsilon\downarrow 0$:

- $\log|B(s_\varepsilon)|\to 0$ uniformly (B is continuous and $|B^*|=1$);
- $\log|\mathcal{O}_\zeta(s_\varepsilon)|\to u(t)$ in L^1_{loc} ($\mathcal{O}_\zeta$ is the Poisson extension of $u:=\log|F^*|$);
- $\log|\zeta(s_\varepsilon)|\to\log|\zeta^*(t)|$ in L^1_{loc} (Lemma 11 or 27);
- $\log|\det_2(s_\varepsilon)|\to\log|\det_2{}^*(t)|$ in L^1_{loc} (BMO boundary trace from the arithmetic Carleson energy, Lemma 24).

Since $u = \log|\det_2{}^*| - \log|\zeta^*|$ by construction of $\mathcal{O}_\zeta$, the sum of boundary traces is $0+u+\log|\zeta^*| - \log|\det_2{}^*| = 0$. Hence $u^*(\frac12+it) = \log|\mathcal{I}^*(t)| = 0$ for a.e. t. *No Smirnov or Hardy class membership is invoked; only the L^1_{loc} convergence of each factor's log-modulus (which the paper proves separately) is needed.*

Growth. $|\mathcal{I}(s)|\leq C(1+|t|)^N$ for some N and all $s=\frac12+\sigma+it$ with $\sigma\in(0,1]$ (this follows from the convexity bound for ζ, the absolutely convergent product for $\det_2$, and the Poisson-controlled modulus of $\mathcal{O}_\zeta$). Hence $u(s) = O(\log(2+|s|)) = o(|s|)$ as $|s|\to\infty$ in Ω.

Conclusion. By the Phragmén–Lindelöf principle for subharmonic functions on the half-plane (e.g. Koosis, *The Logarithmic Integral*, Vol. I, Ch. III; or Ransford [8, Thm. 5.3.4]): a subharmonic function on Ω with nontangential boundary trace ≤ 0 a.e. and growth $o(|s|)$ satisfies $u\leq 0$ on Ω. Hence $|\mathcal{I}|\leq 1$ and $W = -\log|\mathcal{I}|\geq 0$. □

Lemma 32 (Green potential representation for $U = \Re\log\mathcal{J}_{\mathrm{out}}$)**.** *Let $\mathcal{J}_{\mathrm{out}}$ be the outer-normalized unimodular ratio from* (4)*, and set*

$$U(s) \;:=\; \Re\log\mathcal{J}_{\mathrm{out}}(s) \;=\; \log|\mathcal{J}_{\mathrm{out}}(s)| \qquad (s\in\Omega\setminus Z(\zeta)).$$

For $a\in\Omega$, write $a^{\#} := 1-\overline{a}$ for reflection across the boundary line $\Re s=\frac12$, and define the half-plane Green kernel

$$G_\Omega(s,a) \;:=\; \log\left|\frac{s-a^{\#}}{s-a}\right|.$$

Let μ_ξ denote the ξ-zero counting measure in Ω,

$$\mu_\xi \;:=\; \sum_{\rho\in Z(\xi)\cap\Omega} m_\rho\,\delta_\rho,$$

where m_ρ is the multiplicity. Then, in the sense of distributions on Ω,

$$-\Delta\Big(U(s) \;-\; 2\log|B(s)|\Big) \;=\; 2\pi\,\mu_\xi, \qquad B(s) := \frac{s-1}{s}.$$

In particular, up to the explicit compensator $2\log|B|$, the field $U = \Re\log\mathcal{J}_{\mathrm{out}}$ is the Green potential generated by the ξ-zeros in Ω.

Proof. Since $\det_2(I-A)$ and $\mathcal{O}_\zeta$ are holomorphic and nonvanishing on Ω, the poles of $\mathcal{J}_{\text{out}}$ in Ω coincide with the zeros of ζ in Ω, hence with the zeros of ξ in Ω. Moreover, $\mathcal{J}_{\text{out}}$ has a zero at $s=1$ of order 2: one order comes from the simple pole of ζ at $s=1$ (so $1/\zeta$ has a simple zero) and one order comes from the factor $B(s)=(s-1)/s$.

We use the standard divisor identity for a meromorphic function F on a planar domain: $\Delta\log|F| = 2\pi\sum_{\text{zeros}} m\,\delta - 2\pi\sum_{\text{poles}} m\,\delta$ in distributions. Applying this to $\mathcal{J}_{\text{out}}$ gives

$$-\Delta U \;=\; 2\pi\,\mu_\xi \;-\; 4\pi\,\delta_1.$$

Since $B(s)$ is holomorphic on Ω with a simple zero at $s=1$ and no poles in Ω, one has $\Delta\log|B| = 2\pi\,\delta_1$, hence $-\Delta(2\log|B|) = -4\pi\,\delta_1$. Subtracting yields the displayed identity. □

Proposition 33 (Neutralized box-energy bound on Whitney scales)**.** *Let $W=-\log|\mathcal{I}|\ge 0$ be the nonnegative potential from Lemma 31, and let $\widetilde{W} := -\log|B_{\text{far}}\cdot S|$ be the neutralized harmonic field obtained by factoring out the near Blaschke product (see Step 1 below). For each Whitney interval $I=[t_0-L,t_0+L]$ with $L=c/\log\langle t_0\rangle$ and aperture $\alpha'>1$, define the* ***neutralized*** *box energy*

$$E_{\text{neut}}(I) \;:=\; \iint_{Q(\alpha' I)} |\nabla(2\log|B| + \widetilde{W})|^2\,\sigma\,dt\,d\sigma.$$

(This is the energy of $\log|\mathcal{J}_{\text{neut}}|$, the harmonic function on D from the main theorem proof; it does ***not*** *include the infinite-energy near-Blaschke singularities.) Then*

$$E_{\text{neut}}(I) \;\le\; C(\alpha')\,\log^2\langle t_0\rangle\,|I|, \tag{10}$$

*where $C(\alpha')$ depends only on the apertures (α',α''), the RvM density constant, and the convexity exponent—****not on*** *c.*

In particular, the windowed-phase product satisfies

$$\sqrt{E_{\text{neut}}(I)}\cdot L \;\le\; \sqrt{C(\alpha')}\,\frac{c^{3/2}}{\sqrt{\log\langle t_0\rangle}}\,, \tag{11}$$

which tends to 0 as $c\to 0$, uniformly in t_0.

Remark. *The $\log^2\langle t_0\rangle$ growth is* not *an obstruction to the main theorem: in the direct-contradiction proof of Theorem 1, the Whitney parameter is chosen as $c=c_0/\log\langle\gamma_0\rangle$ (depending on the height of the hypothetical zero), which causes $\log^2\langle t_0\rangle\cdot|I|$ to collapse to a height-independent constant $2Cc_0$ (see* (7)*). Replacing $\log^2$ with a uniform constant would allow a fixed c and simplify the argument, but is not logically required.*

Proof. Fix a Whitney interval $I=[t_0-L,t_0+L]$ with $L=c/\log\langle t_0\rangle$ and $\alpha'>1$. Choose a slightly larger aperture parameter $\alpha''>2\alpha'$, and let $D:=Q(\alpha'' I)$ (a dilated Whitney box).

Since $U=2\log|B|+W$ and $B=(s-1)/s$ is explicit and smooth on D (for t_0 large, the clip $L\le L_\star$ keeps D away from $s=1$), $\nabla(2\log|B|)$ contributes $O_{\alpha'}(|I|)$ to the weighted energy. It therefore suffices to bound the W-energy:

$$E_W(I) \;:=\; \iint_{Q(\alpha' I)} |\nabla W|^2\,\sigma\,dt\,d\sigma.$$

Step 1 (Whitney neutralization of $\mathcal{I}$). By Lemma 31, $\mathcal{I}$ is an inner function on Ω with zeros exactly at the nontrivial zeros of ζ in Ω. Factor $\mathcal{I}=e^{i\theta}\,B_{\text{near}}\,B_{\text{far}}\,S$, where B_{near} is the finite Blaschke product over zeros $\rho=\beta+i\gamma$ of $\mathcal{I}$ with $|\gamma-t_0|\le\alpha'' L$, B_{far} is the Blaschke product of the remaining zeros, and S is the (possibly trivial) singular inner factor. By (8), B_{near} has at most $C_{\text{RvM}}(1+2\alpha'' L)\log\langle t_0\rangle = O(\log\langle t_0\rangle)$ factors. (On Whitney scale the count is $O(\log\langle t_0\rangle)$, not $O(1)$; but see Step 2—the near-zero charges do not enter the Cauchy–Schwarz energy bound.)

Define the neutralized field

$$\widetilde{W}(s) \;:=\; W(s)+\log|B_{\text{near}}(s)| \;=\; -\log|B_{\text{far}}(s)| \;-\; \log|S(s)|.$$

Every term on the right is ≥ 0 (each inner factor has modulus ≤ 1 on Ω), so $\widetilde{W} \geq 0$ on Ω. On $\partial\Omega$ ($\sigma = 0$): all inner factors have boundary modulus 1, so $\widetilde{W} = 0$. Moreover, $\widetilde{W}$ is *harmonic* on D: the zeros in B_{far} have $|\gamma - t_0| > \alpha'' L$, hence lie outside the t-span of D, and S is zero-free.

The zeros in B_{near} lie *inside* the box D, so $\log|B_{\text{near}}|$ has logarithmic singularities there and its weighted Dirichlet energy on $Q(\alpha' I)$ is **infinite**. This is not a problem: the near Blaschke factors are absorbed into the **neutralization** step in the main theorem proof (see Step 1 of the proof of Theorem 1), where they cancel the poles of $\mathcal{J}_{\text{out}}$ and produce the harmonic function $\log|\mathcal{J}_{\text{neut}}| = 2\log|B| + \widetilde{W}$ on D. The energy estimate below bounds the harmonic field $\widetilde{W}$ only.

Step 2 (boundary bound for $\widetilde{W}$ on ∂D). Since $\widetilde{W} \geq 0$ and $\widetilde{W} = 0$ on $\sigma = 0$, it remains to bound $\widetilde{W}$ on the top/side edges of D.

Each far zero $\rho = \beta + i\gamma$ with $\delta := \beta - \frac{1}{2} \in (0, \frac{1}{2}]$ contributes

$$-\log|b_\rho(s)| = G_\Omega(s,\rho) = \tfrac{1}{2}\log\frac{(\sigma+\delta)^2 + (t-\gamma)^2}{(\sigma-\delta)^2 + (t-\gamma)^2} \leq \frac{2\sigma\delta}{(\sigma-\delta)^2 + (t-\gamma)^2} \leq \frac{\alpha' L}{(t-\gamma)^2}$$

(using $\log(1+x) \leq x$, $\sigma \leq \alpha' L$, $\delta \leq \frac{1}{2}$, and $|t-\gamma| \geq (\alpha'' - \alpha')L \gg \sigma$). Summing over all far zeros and using the zero density (8) (at most $C_{\text{RvM}}(1+R)\log\langle t_0\rangle$ zeros with $|\gamma - t_0| \leq R$):

$$\sum_{\text{far } \rho} G_\Omega(s,\rho) \;\leq\; \alpha' L \int_{\alpha'' L}^{\infty} \frac{C_{\text{RvM}}\log\langle t_0\rangle}{r^2}\, dr \;=\; \frac{\alpha' C_{\text{RvM}}\log\langle t_0\rangle}{\alpha''} \;\ll\; \log\langle t_0\rangle$$

on ∂D (with the implied constant depending only on α', α'').

Key independence of L and c. The integral $\alpha' L \cdot C_{\text{RvM}}\log\langle t_0\rangle/(\alpha'' L) = \alpha' C_{\text{RvM}}\log\langle t_0\rangle/\alpha''$: the L in the numerator ($\sigma \leq \alpha' L$) cancels the L in the denominator ($\int_{\alpha'' L}^{\infty} 1/r^2\, dr = 1/(\alpha'' L)$). This means the Blaschke tail bound **does not depend on L or c**, and **does not require short-interval zero control** at scale L—only the coarse $O(\log\langle t_0\rangle)$ count per unit ordinate interval.

Singular inner contribution and the $S \equiv 1$ condition. The singular inner factor S of $\mathcal{I}$ contributes $-\log|S(s)| = P_\sigma[\nu_S](t)$, the Poisson integral of a positive singular measure ν_S on $\partial\Omega$. At $\Re s = \frac{3}{2}$: $P_1[\nu_S](t) \leq W(\frac{3}{2} + it) \leq C_0$ (bounded), so ν_S has **uniformly bounded mass per unit interval**: $\nu_S([t_0 - 1, t_0 + 1]) \leq 2\pi C_0 =: \nu_*$.

On ∂D at height $\sigma = \alpha'' L$: the near singular mass ($|\tau - t_0| \leq 1$) contributes at most $\nu_*/(\pi\alpha'' L) = \nu_* \log\langle t_0\rangle/(\pi\alpha'' c)$. **If** $S \equiv 1$ (i.e. $\nu_S = 0$), this vanishes and

$$M \;:=\; \sup_{\partial D} \widetilde{W} \;\leq\; \frac{\alpha' C_{\text{RvM}}}{\alpha''}\,\log\langle t_0\rangle \;=:\; C_*\,\log\langle t_0\rangle,$$

with C_* depending only on $(\alpha', \alpha'', C_{\text{RvM}})$—**not on** c. In this case the energy bound and the contradiction in Theorem 1 close unconditionally (see the remark below).

If $S \not\equiv 1$: the near singular Poisson spike contributes $O(\log\langle t_0\rangle/c)$ to M, which with $c = c_0/\log\langle t_0\rangle$ becomes $O(\log^2/c_0)$ and introduces one extra power of log that the cancellation trick does not absorb. Proving $S \equiv 1$ for the specific inner function $\mathcal{I} = B\mathcal{O}_\zeta\zeta/\det_2$ would therefore complete the unconditional proof; this is recorded as an open step below.

Proof that $S \equiv 1$ (direct $L^1(dt/(1+t^2))$ convergence). The singular inner factor satisfies $S \equiv 1$ if and only if

$$\lim_{\sigma\to 0^+} \int_{\mathbb{R}} \frac{W(\frac{1}{2} + \sigma + it)}{1+t^2}\, dt \;=\; 0$$

(see Garnett [11, Ch. II]). We prove this by showing that each factor of $\mathcal{I} = B\,\mathcal{O}_\zeta\,\zeta/\det_2$ has log-modulus converging in $L^1(\mathbb{R}, dt/(1+t^2))$ as $\sigma \to 0$, and that the boundary traces sum to 0.

Term $\log|B|$. $B = (s-1)/s$ is continuous with $|B^*| = 1$. Convergence is uniform: $\int |\log|B(\sigma)| - 0|/(1+t^2) \to 0$. ✓

Term $\log|\mathcal{O}_\zeta|$. $\mathcal{O}_\zeta$ is the outer function with boundary modulus $\exp(u)$, so $\log|\mathcal{O}_\zeta(\sigma)| = P_\sigma[u] \to u$ in $L^1(dt/(1+t^2))$ (Poisson convergence for $u \in L^1(dt/(1+t^2))$). ✓

Term $\log|\det_2|$. By explicit Fourier computation: $\int \log|\det_2(\sigma,t)|/(1+t^2)\,dt = -\pi\sum_p\sum_{k\ge2} p^{-k(\frac{3}{2}+\sigma)}/k$, which converges absolutely to $-\pi\sum_p\sum_{k\ge2} p^{-3k/2}/k = \int \log|\det_2{}^*|/(1+t^2)\,dt$ as $\sigma\to0$. ✓

Term $\log|\zeta|$ *(the key term).* We must show $\int \log|\zeta(\frac{1}{2}+\sigma+it)|/(1+t^2)\,dt \to \int \log|\zeta^*(t)|/(1+t^2)\,dt$ as $\sigma\to0$.

(a) $\log^+$ *part:* $\log^+|\zeta(\frac{1}{2}+\sigma+it)| \le A\log(2+|t|)$ uniformly for $\sigma\in(0,1]$ (convexity bound; see Titchmarsh, Ch. V). Since $A\log(2+|t|)/(1+t^2)\in L^1$: dominated convergence. ✓

(b) $\log^-$ *part (rigorous majorant via Jensen's inequality):* Fix $R\ge2$ and cover $\mathbb{R}$ by unit intervals $I_n=[n,n+1]$. On each I_n, Jensen's inequality for the *subharmonic* function $\log|\zeta(\frac{1}{2}+\sigma+i\cdot)|$ on a disc of radius 2 centered at $n+\frac{1}{2}+i\sigma$ gives

$$\int_{I_n} \log^-|\zeta(\tfrac{1}{2}+\sigma+it)|\,dt \;\le\; \pi\cdot4\cdot\big(A\log(3+|n|)+C\big) \;+\; \pi\cdot4\cdot N_n\cdot\log4,$$

where N_n is the number of ζ-zeros with $|\gamma-(n+\frac{1}{2})|\le4$ and the right side comes from the standard Jensen bound ($\int\log^-|f| \le$ mean of $\log^+|f|$ on a larger circle + zero count $\cdot\log(\text{ratio})$). By (8): $N_n\le C_1(1+4)\log\langle n\rangle = O(\log\langle n\rangle)$. Hence

$$\int_{I_n} \log^-|\zeta(\sigma,t)|\,dt \;\le\; C_2\log(2+|n|) \qquad \text{uniformly for } \sigma\in(0,1].$$

Dividing by $1+t^2\ge1+n^2$ and summing: $\int_{\mathbb{R}}\log^-|\zeta(\sigma)|/(1+t^2)\le\sum_n C_2\log(2+|n|)/(1+n^2)<\infty$. This bound is **uniform in** σ. ✓

(c) Convergence: L^1_{loc} convergence $\log|\zeta(\sigma)|\to\log|\zeta^*|$ holds by Lemma 27. Combined with the σ-uniform $L^1(dt/(1+t^2))$ bound from (a)+(b), Vitali's convergence theorem (or: L^1_{loc} convergence + uniform integrability of the tail) gives $\int\log|\zeta(\sigma)|/(1+t^2)\to\int\log|\zeta^*|/(1+t^2)$. ✓

Assembly. By the construction of $\mathcal{O}_\zeta$: $u=\log|\det_2{}^*|-\log|\zeta^*|$, so the boundary traces satisfy $0+u+\log|\zeta^*|-\log|\det_2{}^*|=0$. Hence

$$\lim_{\sigma\to0}\int\frac{W(\sigma,t)}{1+t^2}\,dt \;=\; 0-(-u)-(-\log|\zeta^*|)+(-\log|\det_2{}^*|) \;=\; 0.$$

Therefore $S\equiv1$. (This argument uses only: the convexity bound for ζ, the convergence of $\sum 1/(1+\gamma^2)$, the outer construction of $\mathcal{O}_\zeta$, and the explicit Fourier series for $\det_2$. No zero-free hypothesis is used.)

Hence

$$M \;:=\; \sup_{\partial D}\widetilde{W} \;\le\; C_*\,\log\langle t_0\rangle,$$

with C_* independent of c.

Step 3 (interior gradient estimate). Since $\widetilde{W}$ is harmonic on D with $0\le\widetilde{W}\le M$ and $\widetilde{W}=0$ on $\sigma=0$, the standard interior estimate (odd reflection + Cauchy) gives $\sup_{Q(\alpha'I)}|\nabla\widetilde{W}|^2\le C_2M^2/L^2$. Integrating with the weight σ:

$$\iint_{Q(\alpha'I)}|\nabla\widetilde{W}|^2\,\sigma \;\le\; C_3\,M^2\,|I| \;\le\; C_3\,C_*^2\,\log^2\langle t_0\rangle\,|I|.$$

Step 4 (assembly and role of near-zero charges). The energy of the *neutralized harmonic* function $\widetilde{W}$ on $Q(\alpha'I)$ controls the "smooth part" of the boundary phase derivative via the CR–Green pairing (Lemma 37). The $O(\log\langle t_0\rangle)$ zeros of $\mathcal{I}$ inside D contribute *explicit nonnegative charges* $2\pi\sum m_jV_\phi(\rho_j)\ge0$ to the full windowed phase $\int\psi(-w')$ via the distributional Green identity on the punctured domain $D\setminus\{\rho_j\}$. Crucially, these charges **add to the total phase** but do **not enter** the Cauchy–Schwarz energy bound for the smooth part. A hypothetical zero ρ_0 at $\beta_0\ge0.6$ lies **outside** D (since $\delta_0=\beta_0-\frac{1}{2}\ge0.1>\alpha'L$ for t_0 large), so its Poisson contribution enters the smooth part, not the charge term. This is why the contradiction in Theorem 1 is between the

smooth-part lower bound ($\geq 11L$ from ρ_0) and the smooth-part upper bound ($\leq A\sqrt{c_0}\,L$ from the energy of $\widetilde{W}$), independently of the near-zero count.

The effective energy bound for the smooth part is therefore:

$$E_{\text{eff}}(I) \;:=\; \iint_{Q(\alpha' I)} |\nabla \widetilde{W}|^2\,\sigma \;\leq\; C_3\,C_*^2\,\log^2\!\langle t_0\rangle\,|I| \;=:\; C\,\log^2\!\langle t_0\rangle\,|I|,$$

where $C = C_3 C_*^2$ depends only on (α', α'') and is **independent of** c (since C_* depends only on the apertures and the Riemann–von Mangoldt density constant). With $c = c_0/\log\langle t_0\rangle$ in the main theorem: $E_{\text{eff}} = C\log^2 \cdot 2c_0/\log^2 = 2Cc_0$ (height-independent). □

A.6. **From windowed phase control to the wedge.** We convert the established windowed phase bounds into an almost-everywhere wedge inclusion for the boundary values of $\mathcal{J}_{\text{out}}$.

Definition 34 (Admissible window class with atom avoidance)**.** Fix an even C^∞ window ψ with $\psi \equiv 1$ on $[-1,1]$ and $\operatorname{supp}\psi \subset [-2,2]$. For an interval $I = [t_0 - L, t_0 + L]$, an aperture $\alpha' > 1$, and a parameter $\varepsilon \in (0, \frac{1}{4}]$, define $\mathcal{W}_{\text{adm}}(I;\varepsilon)$ to be the set of C^∞, nonnegative, mass-1 bumps ϕ supported in the fixed dilate $2I = [t_0 - 2L, t_0 + 2L]$ that can be written as

$$\phi(t) \;=\; \frac{1}{Z}\frac{1}{L}\,\psi\!\Big(\frac{t-t_0}{L}\Big)\,m(t), \qquad Z = \int_{2I} \frac{1}{L}\psi\!\Big(\frac{t-t_0}{L}\Big)\,m(t)\,dt,$$

where $2I := [t_0 - 2L, t_0 + 2L]$ and the mask $m \in C^\infty(2I;[0,1])$ satisfies:

(i) *Atom avoidance.* There is a union of disjoint open subintervals $E = igcup_{j=1}^{J} J_j \subset I$ with total length $|E| \leq \varepsilon L$ such that $m \equiv 0$ on E and $m \equiv 1$ on $I \setminus E'$, where each transition layer $E' \setminus E$ has thickness $\leq \varepsilon L$.

(ii) *Uniform smoothness.* $\|m'\|_\infty \lesssim (\varepsilon L)^{-1}$ and $\|m''\|_\infty \lesssim (\varepsilon L)^{-2}$ with implicit constants independent of I, t_0, L and of the number/placement of the holes $\{J_j\}$.

Every $\phi \in \mathcal{W}_{\text{adm}}(I;\varepsilon)$ is supported in $2I$. This class contains the unmasked profile $\varphi_{L,t_0}(t) = Z_0^{-1}L^{-1}\psi((t-t_0)/L)$ with $Z_0 := \int_{-2}^{2}\psi(x)\,dx$ (take $E = \varnothing$, $m \equiv 1$) and also allows dodging boundary atoms by punching out small neighborhoods while keeping total deleted length $\leq \varepsilon L$.

Lemma 35 (Uniform Poisson–energy bound for admissible tests)**.** *Let V_ϕ be the Poisson extension of $\phi \in \mathcal{W}_{\text{adm}}(I;\varepsilon)$ to the half-plane, and fix a cutoff to $Q(\alpha' I)$ with $\alpha' > 1$ as in the CR–Green pairing. Then there exists a finite constant $\mathcal{A}_{\text{adm}}(\psi,\varepsilon,\alpha') < \infty$, depending only on $(\psi,\varepsilon,\alpha')$, such that*

$$\iint_{Q(\alpha' I)} |\nabla V_\phi(\sigma,t)|^2\,\sigma\,dt\,d\sigma \;\leq\; \mathcal{A}_{\text{adm}}(\psi,\varepsilon,\alpha')^2\;L.$$

Proof. Let $\phi(t) = Z^{-1}L^{-1}\psi((t-t_0)/L)m(t)$ be an admissible test. By scaling of the Poisson kernel and the uniform bounds on m, m', m'' from Definition 34, the H^1-size of ϕ (equivalently the $L^2(\sigma)$ Dirichlet energy of its Poisson extension on a fixed aperture box) is controlled uniformly by a constant depending only on $(\psi,\varepsilon,\alpha')$, times $L^{1/2}$. Squaring yields the stated $\lesssim L$ energy bound with $\mathcal{A}_{\text{adm}}(\psi,\varepsilon,\alpha')$. □

Lemma 36 (Cutoff pairing on boxes)**.** *Fix parameters $\alpha' > \alpha > 1$. Let $\chi_{L,t_0} \in C_c^\infty(\mathbb{R}^2_+)$ satisfy $\chi \equiv 1$ on $Q(\alpha I)$, $\operatorname{supp}\chi \subset Q(\alpha' I)$, $\|\nabla\chi\|_\infty \lesssim L^{-1}$ and $\|\nabla^2\chi\|_\infty \lesssim L^{-2}$. Let V_ϕ be the Poisson extension of $\phi \in \mathcal{W}_{\text{adm}}(I;\varepsilon)$. Then one has the Green pairing identity*

$$\int_{\mathbb{R}} u(t)\,\phi(t)\,dt \;=\; \iint_{Q(\alpha' I)} \nabla U \cdot \nabla(\chi_{L,t_0}\,V_\phi)\,dt\,d\sigma \;+\; \mathcal{R}_{\text{side}} \;+\; \mathcal{R}_{\text{top}},$$

with remainders satisfying

$$|\mathcal{R}_{\text{side}}| + |\mathcal{R}_{\text{top}}| \;\lesssim\; \Big(\iint_{Q(\alpha' I)} |\nabla U|^2\,\sigma\Big)^{1/2} \cdot \Big(\iint_{Q(\alpha' I)} ig(|\nabla\chi|^2\,|V_\phi|^2 + |\nabla V_\phi|^2 ig)\,\sigma\Big)^{1/2}.$$

Proof. Let $Q := Q(\alpha' I)$. Assume U is C^2 on $\overline{Q}$ and harmonic on Q, with boundary trace $u(t) = U(0,t)$ on the bottom edge $\{\sigma = 0\}$. Since $\chi_{L,t_0} V_\phi$ is compactly supported in $\overline{Q}$ and smooth on Q, Green's identity gives

$$\iint_Q \nabla U \cdot \nabla(\chi V_\phi)\, dt\, d\sigma \;=\; \int_{\partial Q} (\chi V_\phi)\, \partial_n U\, ds \;-\; \iint_Q (\chi V_\phi)\, \Delta U\, dt\, d\sigma.$$

Since $\Delta U = 0$ on Q, only the boundary integral remains. On the bottom edge one has $\partial_n = -\partial_\sigma$, $\chi \equiv 1$, and $V_\phi(0,t) = \phi(t)$, hence that contribution equals

$$\int_I \phi(t)\, (-\partial_\sigma U)(0,t)\, dt.$$

A.7. **Whitney box pairing and local oscillation functional.** We organize the remaining bookkeeping: pairing Whitney boxes across scales and bounding the oscillation functional needed to pass from local control to the global wedge statement.

If U is the real part of a holomorphic logarithm $U = \Re \log J$ with $|J(\frac{1}{2} + it)| = 1$ a.e., then $U(0,t) = 0$ a.e. and $-\partial_\sigma U(0,t) = \partial_t \operatorname{Arg} J(\frac{1}{2} + it)$ in distributions by Cauchy–Riemann; in particular, this term is the tested boundary phase derivative in Lemma 37 below. The remaining boundary pieces (two vertical sides and the top edge) are, by definition, the remainders $\mathcal{R}_{\text{side}} + \mathcal{R}_{\text{top}}$.

For the remainder estimate, we apply Cauchy–Schwarz in the scale-invariant measure $\sigma\, dt\, d\sigma$ on Q:

$$ig|\mathcal{R}_{\text{side}} ig| + ig|\mathcal{R}_{\text{top}} ig| \;\lesssim\; \Big(\iint_Q |\nabla U|^2\, \sigma\Big)^{1/2} \Big(\iint_Q ig|\nabla(\chi V_\phi)|^2\, \sigma\Big)^{1/2}.$$

Expanding $\nabla(\chi V_\phi) = \chi\, \nabla V_\phi + (\nabla \chi)\, V_\phi$ yields

$$\iint_Q ig|\nabla(\chi V_\phi)|^2\, \sigma \;\lesssim\; \iint_Q ig(|\nabla V_\phi|^2 + |\nabla \chi|^2 |V_\phi|^2 ig)\, \sigma,$$

which gives the displayed estimate. □

Lemma 37 (CR–Green pairing for boundary phase)**.** *Let J be analytic on Ω with a.e. boundary modulus $|J(\frac{1}{2} + it)| = 1$, and write $\log J = U + iW$ on Ω, so U is harmonic with $U(\frac{1}{2} + it) = 0$ a.e. Fix a Whitney interval $I = [t_0 - L, t_0 + L]$ and let V_ϕ be the Poisson extension of $\phi \in \mathcal{W}_{\text{adm}}(I; \varepsilon)$. Then, with a cutoff χ_{L,t_0} as in Lemma 36,*

$$\int_{\mathbb{R}} \phi(t)\, ig(-W'(t))\, dt \;=\; \iint_{Q(\alpha' I)} \nabla U \cdot \nabla(\chi_{L,t_0}\, V_\phi)\, dt\, d\sigma \;+\; \mathcal{R}_{\text{side}} \;+\; \mathcal{R}_{\text{top}},$$

and the remainders satisfy the same estimate as in Lemma 36. In particular, by Cauchy–Schwarz and Lemma 35, there is a constant $C_{\text{rem}}(\alpha', \psi)$ such that

$$\int_{\mathbb{R}} \phi(t)\, ig(-w'(t))\, dt \;\le\; C_{\text{rem}}(\alpha', \psi) \Big(\iint_{Q(\alpha' I)} |\nabla U|^2\, \sigma\Big)^{1/2}.$$

Proof. On the bottom edge $\{\sigma = 0\}$ the outward normal is $\partial_n = -\partial_\sigma$. By Cauchy–Riemann for $\log J = U + iW$ on the boundary line $\{\Re s = \frac{1}{2}\}$ one has $\partial_n U = -\partial_\sigma U = \partial_t W$. Thus the bottom-edge term in Green's identity is

$$-\int_{\partial Q \cap \{\sigma = 0\}} \chi\, V_\phi\, \partial_n U\, dt = -\int_{\mathbb{R}} \phi(t)\, \partial_t W(t)\, dt = \int_{\mathbb{R}} \phi(t)\, ig(-w'(t))\, dt,$$

which yields the stated identity after including the interior term and remainders. The final inequality is Cauchy–Schwarz together with the uniform Poisson-energy bound from Lemma 35. □

Proposition 38 (Length-independent upper bound for admissible tests)**.** *Let J be holomorphic on $\Omega \setminus Z(\zeta)$ with a.e. boundary modulus 1, write $\log J = U + iW$ on $\Omega \setminus Z(\zeta)$, and let $-w'$ denote*

the boundary phase distribution. For every interval $I = [t_0 - L, t_0 + L]$, every $\phi \in \mathcal{W}_{\mathrm{adm}}(I;\varepsilon)$, and every fixed cutoff to $Q(\alpha' I)$,

$$\int_{\mathbb{R}} \phi(t)\,(-w')(t)\,dt \;\le\; C_{\mathrm{test}}(\psi,\varepsilon,\alpha')\Big(\iint_{Q(\alpha' I)} |\nabla U|^2\,\sigma\,dt\,d\sigma\Big)^{1/2} \tag{12}$$

with $C_{\mathrm{test}}(\psi,\varepsilon,\alpha') := C_{\mathrm{rem}}(\alpha',\psi)\,\mathcal{A}_{\mathrm{adm}}(\psi,\varepsilon,\alpha')$ independent of I, t_0, L.

Proof. Apply Lemma 37 with $\phi \in \mathcal{W}_{\mathrm{adm}}(I;\varepsilon)$ and absorb the window-side constants into $C_{\mathrm{test}}(\psi,\varepsilon,\alpha')$. □

Lemma 39 (Whitney–uniform wedge bound)**.** *Fix parameters $\alpha' > 1$ and $\varepsilon \in (0, \frac{1}{4}]$. Fix the Whitney schedule: set*

$$L(t_0) \;:=\; \frac{c}{\log\langle t_0\rangle}, \qquad c \in (0,1].$$

Then for every Whitney interval $I = [t_0 - L, t_0 + L]$ and the corresponding cutoff $\psi_{L,t_0}(t) := \psi((t-t_0)/L) = Z_0 L\,\varphi_{L,t_0}(t)$ (so $\psi_{L,t_0} \equiv 1$ on I), the windowed phase satisfies the uniform bound

$$\int_{\mathbb{R}} \psi_{L,t_0}(t)\,(-w'(t))\,dt \;\le\; Z_0\,C_{\mathrm{test}}(\psi,\varepsilon,\alpha')\sqrt{2C_{\mathrm{box}}(\alpha',c)}\,L^{3/2} \;\le\; Z_0\,C_{\mathrm{test}}\sqrt{2C}\,\frac{c^{3/2}}{\sqrt{\log 2}} \;=:\; \pi\,\Upsilon_{\mathrm{Whit}}(c).$$

Choosing $c > 0$ sufficiently small makes $\Upsilon_{\mathrm{Whit}}(c) < \frac{1}{2}$, bounding the windowed phase variation on every Whitney interval. (A separate local-to-global argument—not supplied in this version—would be needed to promote this windowed bound to the full boundary wedge (P+)*.)*

Proof. Since $\psi_{L,t_0} = Z_0 L\,\varphi_{L,t_0}$, Proposition 38 gives

$$\int \psi_{L,t_0}\,(-w') \;=\; Z_0 L \int \varphi_{L,t_0}\,(-w') \;\le\; Z_0 L \cdot C_{\mathrm{test}}\sqrt{E(I)}\,.$$

By the bound (11) of Proposition 33, $\sqrt{E(I)} \cdot L \le \sqrt{2C}\,c^{3/2}/\sqrt{\log\langle t_0\rangle}$, so

$$Z_0 L \cdot C_{\mathrm{test}}\sqrt{E(I)} \;=\; Z_0 C_{\mathrm{test}} \cdot \sqrt{E(I)} \cdot L \;\le\; Z_0 C_{\mathrm{test}}\sqrt{2C}\,\frac{c^{3/2}}{\sqrt{\log 2}}\,,$$

which is uniformly bounded in t_0 and tends to 0 as $c \to 0$. □

Remark 40 (Energy-budget mechanism underlying (P+)). The mechanism that drives (P+) admits a transparent decomposition into "source cost" and "field cost" (cf. the explicit-formula duality between primes and zeros).

Source (prime) side. The regularized determinant $\det_2(I - A(s))$ is holomorphic and *zero-free* on Ω (Lemma 2); it contributes the arithmetic Carleson constant $K_0 = \frac{1}{4}\sum_p \sum_{k\ge 2} p^{-k}/k^2 < \infty$ (Lemma 24). This constant is unconditionally finite because the prime series converges on the half-plane—it encodes the total "cost" of the irreducible multiplicative generators.

Field (zero) side. The zeros of ζ in Ω contribute to $|\nabla U|^2$ through the Green-potential representation (Lemma 32). After neutralizing the $O(1)$ near-field zeros by a local Blaschke product, the harmonic residual $\widetilde{U}$ satisfies a boundary bound $|\widetilde{U}| \le M = O(\log\langle T\rangle)$ on the dilated Whitney box, by the Poisson-kernel averaging of the far-field Green contributions together with the short-interval Riemann–von Mangoldt density estimate. The interior gradient estimate for harmonic functions then gives a box energy $E(I) \le C\log^2\langle T\rangle \cdot |I|$ that may grow with height but is *overwhelmed* by the Whitney scaling $L^{3/2} = (c/\log\langle T\rangle)^{3/2}$.

The decisive algebraic fact is that the product $\sqrt{E(I)} \cdot L = O(c^{3/2}/\sqrt{\log\langle T\rangle})$ tends to 0 as $c \to 0$ *uniformly in height*. The Whitney–uniform wedge (Lemma 39) converts this vanishing product into a phase bound $\Upsilon_{\mathrm{Whit}}(c) < \frac{1}{2}$, and the local-to-global lemma delivers (P+). In short: bounded near-field cost + Poisson-averaged far-field control ⇒ height-dependent box energy beaten by Whitney shrinkage ⇒ small windowed phase ⇒ boundary wedge.

Remark 41 (Status of the (P+) route). The boundary wedge (P+) has *not* been established unconditionally in this version. Lemma 39 bounds the windowed phase variation on each Whitney interval by $\pi\Upsilon_{\mathrm{Whit}}(c)$, which can be made $< \pi/2$ by choosing c small. However, a separate *local-to-global* argument (promoting windowed control to the full Lebesgue-a.e. wedge inclusion) is required but not supplied here. Moreover, the (P+) statement as written (a single global wedge for all heights) would, combined with the Smirnov transport in §4, rule out zeros for *all* $\Re s > 1/2$—which is far stronger than the claimed $\Re s \geq 0.6$ result and is not what the current analysis establishes.

The direct-contradiction proof of Theorem 1 in §4.1 is the intended primary route. Proposition 33 gives the energy bound $E(I) \leq C \log^2\langle t_0\rangle\, |I|$ (height-dependent, with $\log^2$ growth). The direct contradiction closes by choosing the Whitney parameter $c = c_0/\log\langle\gamma_0\rangle$, which causes the $\log^2$ factor to cancel against $|I| \propto 1/\log^2$, yielding a height-independent ratio $A\sqrt{c_0}/11 < 1$ for $c_0 < (11/A)^2/2$.

Appendix B. Supplementary computational verification protocol (verifier commands and expected fields)

There are two verification modes:

- **Verification using shipped artifacts:** verify the shipped JSON artifacts match Table 1.
- **Optional regeneration check (supplementary):** rerun the verifier to regenerate the artifacts from scratch.

Prerequisites. From the bundle's `compute/` directory, install the ARB/ball-arithmetic bindings:

```
pip install -r requirements.txt
```

Verification using shipped artifacts: check shipped JSON artifacts.

- **Rectangle artifact** `artifacts/theta_certify_sigma06_07_t0_20_outer_zeta_proj.json`. Check (at minimum):
  - `results.ok = true`
  - `results.theta_hi = 0.9999928763...  < 1`
  - `results.processed_boxes = 380764`
- **Pick artifact** `artifacts/pick_sigma0599_raw_zeta_N16.json`. Check (at minimum):
  - `pick.delta_cert = 0.594...`
  - `pick.P_spd_at_0 = true`
  - `pick.tail_l1_partial_hi` (diagnostic L1 tail sum)
- **Pick artifact** `artifacts/pick_sigma06_raw_zeta_N16.json`. Check (at minimum):
  - `pick.delta_cert = 0.594...`
  - `pick.P_spd_at_0 = true`
  - `pick.tail_l1_partial_hi` (diagnostic L1 tail sum)
- **Pick artifact** `artifacts/pick_sigma07_raw_zeta_N16.json`. Check (at minimum):
  - `pick.delta_cert = 0.627...`
  - `pick.P_spd_at_0 = true`
  - `pick.tail_l1_partial_hi` (diagnostic L1 tail sum)

Optional regeneration check (supplementary): exact command lines. Run the verifier from the bundle's `compute/` directory (or the corresponding directory in a repository checkout). The following commands reproduce the primary artifacts (line breaks are for readability):
1) Rectangle certification (`theta_certify`).

```
python verify_attachment_arb.py \
  --theta-certify \
  --arith-gauge outer_zeta_proj \
  --arith-P-cut 2000 \
  --rect-sigma-min 0.6 --rect-sigma-max 0.7 \
```

```
  --rect-t-min 0.0 --rect-t-max 20.0 \
  --outer-mode midpoint \
  --outer-P-cut 2000 \
  --outer-T 50.0 --outer-n 2001 \
  --theta-init-n-sigma 10 --theta-init-n-t 50 \
  --theta-min-sigma-width 0.0001 --theta-min-t-width 0.001 \
  --theta-max-boxes 500000 \
  --prec 260 \
  --theta-out artifacts/theta_certify_sigma06_07_t0_20_outer_zeta_proj.json \
  --progress
```

2) Pick certification at $\sigma_0 = 0.599$ (`pick_certify`).

```
python verify_attachment_arb.py \
  --pick-certify \
  --pick-sigma0 0.599 \
  --pick-N 16 \
  --pick-coeff-count 128 \
  --pick-K 512 \
  --pick-rho 0.4 \
  --pick-rho-bound 0.5 \
  --arith-gauge raw_zeta \
  --arith-P-cut 2000 \
  --prec 1024 \
  --pick-out artifacts/pick_sigma0599_raw_zeta_N16.json
```

3) Pick certification at $\sigma_0 = 0.6$ (`pick_certify`).

```
python verify_attachment_arb.py \
  --pick-certify \
  --pick-sigma0 0.6 \
  --pick-N 16 \
  --pick-coeff-count 128 \
  --pick-K 512 \
  --pick-rho 0.4 \
  --pick-rho-bound 0.5 \
  --arith-gauge raw_zeta \
  --arith-P-cut 2000 \
  --prec 1024 \
  --pick-out artifacts/pick_sigma06_raw_zeta_N16.json
```

4) Pick certification at $\sigma_0 = 0.7$ (`pick_certify`).

```
python verify_attachment_arb.py \
  --pick-certify \
  --pick-sigma0 0.7 \
  --pick-N 16 \
  --pick-coeff-count 128 \
  --pick-K 512 \
  --pick-rho 0.4 \
  --pick-rho-bound 0.5 \
  --arith-gauge raw_zeta \
  --arith-P-cut 2000 \
  --outer-mode rigorous \
  --outer-P-cut 2000 \
```

```
  --prec 1024 \
  --pick-out artifacts/pick_sigma07_raw_zeta_N16.json
```

Meaning of successful verification. The verifier uses *ball arithmetic*: each computed quantity is an interval enclosure (midpoint plus radius) and every operation propagates rounding error outward. Thus each check is a formal inequality of the form `upper bound < 1` or `directed-rounding` $\mathrm{LDL}^\top$ `succeeds with positive pivots`. If the verification checks above pass, then the numerical inequalities summarized in Table 1 are within the logic of ball arithmetic.

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